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I've been working on this research paper for several months now. What are its fatal weaknessess ? how can i complete it into the platonic ideal shape it yearns to be, guided by its own inner music? what are the top 1000 literal line item fixes I have to make, that altogether UPGRADE the mathematics and the physics? never downgrade? Only tolerate realizing the strongest possible version of this paper imaginable/conceivable.

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Thought for 7m 55s >

According to a document from 2026-06-05, the manuscript already contains many of the fragments needed for a serious monograph, but they are not yet assembled at the level of *construction*. The defect is not only "missing proof." It is that the draft repeatedly states objects before giving their local models, residues, signs, propagators, obstruction complexes, comparison maps, and analytic estimates. The right repair is not another architecture paragraph; it is to replace slogans by the following pieces of mathematics. main

1. The chain object must be written, not named

The two-coloured object is not a metaphor. It is the following dg coloured dioperad.

Let the colours be c, o. For $p, q \geq 0$,

$$SC_{\mathrm{ch,top}}(c^p; c) = C_\bullet^{\log}\big(FM_p(\mathbb{C})\big),$$

$$SC_{\mathrm{ch,top}}(c^p, o^q; o) = C_\bullet^{\log}\big(FM_p(\mathbb{C}) \times \mathrm{Conf}_q^<(\mathbb{R})\big),$$

$$SC_{\mathrm{ch,top}}(c^p, o^q; c) = 0 \quad \text{if } q > 0.$$

The composition is

$$\gamma_c : FM_p(\mathbb{C}) \times \prod_{i=1}^{p} FM_{n_i}(\mathbb{C}) \longrightarrow FM_{n_1+\cdots+n_p}(\mathbb{C}),$$

given by Fulton–MacPherson insertion of infinitesimal configurations into marked points, and

$$\gamma_o : \big(FM_p(\mathbb{C}) \times \mathrm{Conf}^{<}_q(\mathbb{R})\big) \times \prod_i FM_{n_i}(\mathbb{C}) \times \prod_j \mathrm{Conf}^{<}_{m_j}(\mathbb{R}) \to FM_{p+\sum n_i}(\mathbb{C}) \times \mathrm{Conf}^{<}_{q+\sum m}$$

with closed inputs inserted into closed slots and open intervals inserted into open slots. The no-open-to-closed component is a vanishing statement, not prose:

$$C^{\log}_{\bullet}\big(SC_{\mathrm{ch,top}}(c^p, o^q; c)\big) = 0, \qquad q > 0.$$

The manuscript says this directionality is forced geometrically, at the D-module support level, and by the factorized HT parametrix; that must become the lemma proving the zero component. main

The W-resolution is then

$$W(SC_{\mathrm{ch,top}})(a; b) = \bigoplus_T \mathfrak{o}(T) \otimes \bigotimes_{v \in V(T)} SC_{\mathrm{ch,top}}(\mathrm{in}(v); \mathrm{out}(v)) \otimes C_{\bullet}([0,1])^{\otimes E_{\mathrm{int}}(T)},$$

with differential

$$d_W = d_{\mathrm{chain}} + d_{[0,1]} + \sum_{e:\, s_e = 0} \pm\, \mathrm{contract}_e + \sum_{e:\, s_e = 1} \pm\, \mathrm{compose}_e\,.$$

The missing line is the sign convention

$$\mathfrak{o}(T) = \det(E_{\mathrm{int}}(T)) \otimes \bigotimes_{v \in V(T)} \mathfrak{o}(v),$$

with the edge-ordering sign fixed once and for all. Without this orientation line, the two-coloured coherence theorem is not a theorem.

2. The A_∞-chiral algebra must be given as a coderivation

The concrete object is

$$B_{\mathrm{ch}}(A) = T^c(s^{-1}\bar{A}) = \bigoplus_{n \ge 0} (s^{-1}\bar{A})^{\otimes n}.$$

For homogeneous a_i, put

$$\bar{a}_i := |a_i| - 1.$$

A degree $+1$ coderivation D_A is determined by maps

$$m_k : A^{\otimes k} \to A((\lambda_1)) \cdots ((\lambda_{k-1})), \qquad |m_k| = 2 - k,$$

through

$$D_A[s^{-1}a_1|\cdots|s^{-1}a_n] = \sum_{\substack{1\le i\le n-k+1\\ k\ge 1}} (-1)^{\epsilon(i,k)}[s^{-1}a_1|\cdots|s^{-1}m_k(a_i,\ldots,a_{i+k-1})|\cdots|s^{-1}a_n],$$

where

$$\epsilon(i,k) = \sum_{j<i} \bar{a}_j$$

in the suspended convention. If a second convention is used, the paper must write the suspension isomorphism

$$m_k = s \circ m_k \circ (s^{-1})^{\otimes k}$$

and recompute $\epsilon(i,k)$. The manuscript already records that the m_k's are the cogenerator projections of a degree $+1$ coderivation, but this formula must become the first displayed formula of the A_∞-chiral chapter. main

The coalgebra condition is

$$\Delta D_A = (D_A \otimes 1 + 1 \otimes D_A)\Delta,$$

and the whole A_∞-chiral structure is the single equation

$$D_A^2 = 0.$$

Equivalently,

$$\sum_{\substack{r+s+t=n\\ s\ge 1}} (-1)^{r+st} m_{r+1+t}\big(a_1,\ldots,a_r,m_s(a_{r+1},\ldots,a_{r+s}),a_{r+s+1},\ldots,a_n\big) = 0,$$

with spectral substitution imposed by the collapsed interval. For consecutive variables

$$\lambda_i = z_i - z_{i+1},$$

the effective variable for a collapsed block $a_{r+1},\ldots,a_{r+s}$ is

$$\Lambda_{r+1,r+s} = \lambda_{r+1} + \cdots + \lambda_{r+s-1}.$$

The outer operation sees the collapsed point as one input. Thus the outer consecutive variables are

$$\lambda'_j = \begin{cases} \lambda_j, & j<r,\\ \lambda_r + \lambda_{r+1} + \cdots + \lambda_{r+s-1}, & j=r,\\ \lambda_{j+s-1}, & j>r. \end{cases}$$

This substitution law is not cosmetic. It is the algebraic image of the boundary face

$$D_{\{r+1,\ldots,r+s\}} \subset \partial FM_n(\mathbb{C}).$$

The manuscript has the right material around the concrete axioms and the operad-to-m_k construction; it needs to promote the spectral substitution law to the main identity, not a convention. main

3. Sesquilinearity must be derived from translation equivariance

The correct formulas are

$$m_k(\ldots,\partial a_i,\ldots) = -\big(\lambda_i+\cdots+\lambda_{k-1}\big)m_k(\ldots,a_i,\ldots), \qquad 1\le i<k,$$

and

$$m_k(a_1,\ldots,\partial a_k) = \left(\partial_A + \sum_{j=1}^{k-1} j\lambda_j\right) m_k(a_1,\ldots,a_k).$$

The second formula is the place where many drafts silently lose a factor. The coefficient $j\lambda_j$ comes from

$$\sum_{i=1}^{k-1}\lambda_i^{\text{indep}} = \sum_{i=1}^{k-1}\sum_{j=i}^{k-1}\lambda_j^{\text{consec}} = \sum_{j=1}^{k-1} j\lambda_j^{\text{consec}}.$$

The unit identities are

$$m_1(1)=0, \qquad \partial 1 = 0,$$

$$m_2(1,a)=a, \qquad m_2(a,1)=(-1)^{|a|}a,$$

$$m_k(\ldots,1,\ldots)=0, \qquad k\neq 2.$$

These formulas are present in the manuscript; they need to be used everywhere.

main

4. The Fulton–MacPherson residue calculus must control the signs

For a collision divisor

$$D_S\subset FM_k(\mathbb{C}), \qquad |S|\ge 2,$$

choose an inward radial parameter

$$\varepsilon_S \ge 0$$

and local logarithmic normal form

$$d\log\varepsilon_S.$$

The orientation convention must be

$$\operatorname{or}(FM_k(\mathbb{C})) = (-d\varepsilon_S) \wedge \operatorname{or}(D_S).$$

For

$$\omega \in \Omega^{k-1}_{\log}(FM_k(\mathbb{C})),$$

write near D_S

$$\omega = d\log\varepsilon_S \wedge \beta + \alpha,$$

with α, β smooth along the boundary. Then

$$\operatorname{Res}_{D_S}(\omega) = \beta|_{\varepsilon_S=0}.$$

Equivalently,

$$\operatorname{Res}_{D_S}(\omega) = \lim_{\varepsilon_S \to 0} \varepsilon_S \, \iota_{\partial_{\varepsilon_S}} \omega \; {}_{D_S}.$$

The manuscript explicitly warns that the naive contraction without the factor ε_S diverges; this must become the default residue definition. main

Now define

$$D_{\text{coll}} = \sum_S (-1)^{\epsilon_S} \operatorname{Res}_{D_S} \otimes m_S.$$

The proof of $D^2_{\text{coll}} = 0$ is not "associahedra." It is

$$\partial^2[FM_k(\mathbb{C})] = 0.$$

At a codimension-two corner $D_S \cap D_T$, one has two iterated residues:

$$\operatorname{Res}_{D_T} \operatorname{Res}_{D_S}(\omega) \quad \text{and} \quad \operatorname{Res}_{D_S} \operatorname{Res}_{D_T}(\omega).$$

The two ordered approaches carry opposite incidence signs because

$$d\varepsilon_T \wedge d\varepsilon_S = -d\varepsilon_S \wedge d\varepsilon_T.$$

Thus

$$\int_{(D_S)|D_T} \operatorname{Res}_{D_S} \operatorname{Res}_{D_T} \omega + \int_{(D_T)|D_S} \operatorname{Res}_{D_T} \operatorname{Res}_{D_S} \omega = 0.$$

This is the local form of

$$\sum_{S \neq T} \epsilon_S \epsilon_T [D_S \cap D_T] = 0.$$

The manuscript contains exactly this orientation mechanism; it needs to be placed before every low-arity computation. main

5. The Arnold relation must be the Jacobi identity, not a slogan

On $\mathrm{Conf}_n(\mathbb{C})$, put

$$\omega_{ij} = d\log(z_i - z_j) = \frac{dz_i - dz_j}{z_i - z_j}.$$

For $i < j < k$,

$$\omega_{ij} \wedge \omega_{jk} + \omega_{jk} \wedge \omega_{ki} + \omega_{ki} \wedge \omega_{ij} = 0.$$

At $n = 3$, this identity is the Jacobi identity for the singular part of m_2. At $n = 4$, it is the cancellation of the two ways of reaching a codimension-two corner of $FM_4(\mathbb{C})$. The manuscript's discussion of FM_4 says the nine nonvanishing terms are supported only on consecutive boundary strata, while nonconsecutive divisors vanish by the ordering constraint; this should be the printed proof of the $n = 4$ relation, not a verbal statement. main

The minimal-case correction is essential:

If $m_1 = 0$, the arity-four A_∞ identity is

$$m_2 \circ m_3 + m_3 \circ m_2 = 0.$$

It does **not** determine m_4. The first identity in which m_4 appears is arity five:

$$m_2 \circ m_4 + m_3 \circ m_3 + m_4 \circ m_2 = 0.$$

Therefore every formula in the manuscript deriving m_4 from the $n = 4$ Stasheff identity must be corrected to an arity-five obstruction computation.

6. The chiral Hochschild complex must be written as coderivations

The bulk is

$$Z^{\mathrm{der}}_{\mathrm{ch}}(A) = C^\bullet_{\mathrm{ch}}(A, A) \simeq \mathrm{Coder}(B_{\mathrm{ch}}(A)),$$

with differential

$$d_{\mathrm{Hoch}} f = [D_A, f].$$

A coderivation f of arity p has corestriction

$$f_p : (s^{-1}\bar{A})^{\otimes p} \to s^{-1}\bar{A}.$$

The brace is

$$(f\{g_1, \ldots, g_r\})_n = \sum (-1)^\epsilon f_{n-r+\sum p_i}\big(1^{\otimes i_1}, g_{1,p_1}, 1^{\otimes i_2}, \ldots, g_{r,p_r}, 1^{\otimes i_{r+1}}\big),$$

where the sum runs over ordered insertions

$$0 \le i_1 \le i_1 + p_1 \le i_2 \le \cdots \le n.$$

The sign is the Koszul sign obtained by moving g_j of suspended degree $|g_j| - 1$ past the preceding suspended inputs:

$$\epsilon = \sum_{j=1}^{r} (|g_j| - 1) \sum_{\ell < \mathrm{pos}(g_j)} \bar{a}_\ell \quad + \sum_{i<j} (|g_i| - 1)(p_j - 1).$$

The Gerstenhaber bracket is

$$[f, g] = f\{g\} - (-1)^{(|f|-1)(|g|-1)} g\{f\}.$$

The mixed open/closed operations must be explicitly

$$\nu_k^{(m)} : \mathrm{Coder}(B_{\mathrm{ch}}(A))^{\otimes k} \otimes A^{\otimes m} \to A,$$

$$\nu_k^{(m)}(F_1, \ldots, F_k; a_1, \ldots, a_m) = \sum_{\Gamma \in \mathrm{Ins}(k,m)} (-1)^{\epsilon(\Gamma)} \, \mathrm{ev}_A\big(F_1, \ldots, F_k \text{ inserted in the } A\text{-word } a_1 | \cdots$$

The manuscript says these mixed operations are the chiral brace insertions $\nu_k^{(m)}$ and that the two-coloured assembly requires an extra cobar homotopy. That should be promoted into a definition-proposition-proof block. main

7. The missing h_{oc} must be a formula on trees

The missing object is not "some coherence." It is a contracting homotopy

$$h_{\mathrm{oc}} : \Omega\big((SC_{\mathrm{ch,top}})^!\big) \to \Omega\big((SC_{\mathrm{ch,top}})^!\big)[-1]$$

satisfying

$$dh_{\mathrm{oc}} + h_{\mathrm{oc}} d = \mathrm{id} - ip.$$

It must satisfy

$$h_{\mathrm{oc}}\big(SC(c^p, o^q; c)\big) = 0 \qquad (q > 0),$$

and

$$\Delta_{\mathrm{mix}} h_{\mathrm{oc}} = (h_{\mathrm{oc}} \otimes 1 + 1 \otimes h_{\mathrm{oc}})\Delta_{\mathrm{mix}}$$

up to the specified Koszul sign.

Concretely, on a two-coloured tree T, define

$$h_{\mathrm{oc}}(T) = \sum_{e \in E_{\mathrm{int}}(T)} (-1)^{\sigma(e,T)} T_e,$$

where T_e is obtained by inserting a Boardman–Vogt edge length parameter and expanding the edge e into the corresponding cobar generator. The sign must be

$$\sigma(e,T) = \#\{e' < e\} + \sum_{v \prec e} |\operatorname{dec}(v)|$$

for a fixed total ordering of internal edges and vertex decorations. Then one verifies

$$dh_{\mathrm{oc}}(T) + h_{\mathrm{oc}}d(T) = T - ip(T)$$

by the cancellation of all twice-expanded edges. This is the two-coloured analogue of the usual bar-cobar contraction; the manuscript correctly says this two-coloured lift is not established at chain level. main

The obstruction class is

$$[o_{\mathrm{oc}}] \in H^1 \operatorname{Cone}\left(\Omega((SC_{\mathrm{ch,top}})^!) \to \operatorname{End}_{(Z,A)}\right).$$

Until this class is killed, the paper has E_2-braces and a cohomological two-coloured statement, not a strict chain-level Swiss-cheese theorem.

8. The E_3-PBW route must be a theorem, not a hope

The missing theorem is

$$Z^{\mathrm{der}}_{\mathrm{ch}}(A|_{D_x}) \simeq U_{E_3}(W_x),$$

where W_x is a polynomial-growth graded Lie object controlling the local chiral tangent complex. The associated graded must be

$$\operatorname{gr} Z^{\mathrm{der}}_{\mathrm{ch}}(A|_{D_x}) \simeq \operatorname{Sym}_{E_3}\left(W_x[-2]\right).$$

Then the spectral sequence is

$$E_1^{p,q} = \operatorname{Sym}^p_{E_3} H^q(W_x[-2]) \Longrightarrow H^{p+q} Z^{\mathrm{der}}_{\mathrm{ch}}(A|_{D_x}).$$

Collapse requires a weight estimate

$$\dim W_{x,w} \leq Cw^N,$$

and a vanishing region

$$H^q(W_x) = 0 \qquad q > N_0.$$

Your manuscript explicitly names this as an unproved proof obligation: the chiral-E_3-PBW theorem presenting the local derived centre as $U_{E_3}(W_x)$ has not been established. main

9. The BV construction must be written with fields, kernels, and QME

The 3d HT bridge must start with a BV datum

$$(\mathcal{E}, Q, \omega_{BV}, I, \mathcal{B}),$$

where

$$\mathcal{E} = \Omega^\bullet(\mathbb{R}_t) \otimes \Omega^{0,\bullet}(\mathbb{C}_z) \otimes \mathfrak{a}[1],$$

$$Q = d_t + \bar{\partial}_z + d_{\mathfrak{a}},$$

and

$$\omega_{BV}(\alpha, \beta) = \int_{\mathbb{R}\times\mathbb{C}} dz \wedge \langle \alpha, \beta \rangle_{\mathfrak{a}}.$$

For a cyclic dg Lie algebra $\mathfrak{a}$, the classical action is

$$S_{\mathrm{cl}}[A] = \int_{\mathbb{R}\times\mathbb{C}} dz \wedge \left(\frac{1}{2} \langle A, QA \rangle + \frac{1}{6} \langle A, [A, A] \rangle \right).$$

Then

$$\{S_{\mathrm{cl}}, S_{\mathrm{cl}}\}_{BV} = 0$$

by invariance of the pairing and Jacobi.

At scale L, choose a heat kernel

$$K_s = e^{-sQQ^*}$$

and propagator

$$P_{\epsilon<L} = \int_\epsilon^L Q^* K_s \, ds.$$

The effective interaction is

$$I[L] = \sum_\Gamma \frac{\hbar^{b_1(\Gamma)}}{|\operatorname{Aut}\Gamma|} W_\Gamma(P_{0<L}, I),$$

with

$$W_\Gamma(P, I) = \int_{\mathrm{Conf}_{V(\Gamma)}(\mathbb{R}\times\mathbb{C})} \bigwedge_{e\in E(\Gamma)} P_e \cdot \bigotimes_{v\in V(\Gamma)} I_v.$$

The quantum master equation is

$$QI[L] + \hbar\Delta_L I[L] + \frac{1}{2}\{I[L], I[L]\}_L = 0.$$

The boundary A_∞-chiral operations are extracted by restricting external legs to the boundary condition $\mathcal{B}$:

$$m_k^{\mathrm{BV}} = \sum_{\substack{\Gamma \text{ connected} \\ k \text{ external boundary legs} \\ 1 \text{ output}}} \frac{\hbar^{b_1(\Gamma)}}{|\operatorname{Aut}\Gamma|} \operatorname{Res}_{\partial FM_\Gamma} W_\Gamma(P, I).$$

This is the physical equality that must match

$$m_k^{\mathrm{bar}} = \pi_1 D_A|_{(s^{-1}A)^{\otimes k}}.$$

The manuscript has the Costello–Wilson model in this form for the $6d$ holomorphic Chern–Simons source: fields

$$A \in \Omega^{0,\bullet}(K3 \times E) \otimes \Omega^\bullet(\mathbb{R}) \otimes \mathfrak{a}_{\mathrm{hCS}}[1],$$

operator

$$Q = d_{\mathbb{R}} + \bar{\partial}_{K3} + \bar{\partial}_E,$$

and action

$$S_{\mathrm{cl}}[A] = \int_{\mathbb{R}\times K3\times E} \Omega_{K3} \wedge \Omega_E \wedge \left(\frac{1}{2}\langle A, QA\rangle + \frac{1}{6}\langle A, [A, A]\rangle \right).$$

main

This is the level of field-theoretic detail every physical realization section needs.

10. The Heisenberg test must be fully normalized

For the rank-one Heisenberg VOA H_k, with generator J,

$$J(z)J(w) \sim \frac{k}{(z-w)^2}.$$

Then

$$\{J_\lambda J\} = k\lambda.$$

The bar differential is

$$D[J|J] = k,$$

after the suspension convention is fixed. Higher operations vanish:

$$m_k = 0, \qquad k \geq 3.$$

The collision residue is

$$r_{H_k}(z) = \frac{k}{z}.$$

The quantum line braiding on Fock lines of charges μ_1, μ_2 is

$$R_{\mu_1,\mu_2}(z) = \exp\left(\frac{\hbar k \mu_1 \mu_2}{z}\right).$$

The genus scalar is

$$F_g(H_k) = k\,\lambda_g^{FP}.$$

The genus-one trace is

$$F_1(H_k) = \frac{k}{24}.$$

This is the benchmark. Every normalization for affine, Virasoro, W_3, and W_N should be reduced to this calculation.

11. The affine test must distinguish trace-form and KZ normalizations

For affine $V_k(\mathfrak{g})$,

$$J^a(z)J^b(w) \sim \frac{k(a,b)}{(z-w)^2} + \frac{f^{ab}_{\ c}J^c(w)}{z-w}.$$

Thus

$$\{J^a_\lambda J^b\} = f^{ab}_{\ c}J^c + k(a,b)\lambda.$$

The trace-form collision residue is

$$r_{\mathrm{tr}}(z) = \frac{k\,\Omega}{z}.$$

The KZ normalization is

$$r_{\mathrm{KZ}}(z) = \frac{\Omega}{(k+h^\vee)z}.$$

They are not the same object; they are related by the Sugawara/KZ comparison, not by notation. The KZ connection is

$$\nabla_{\mathrm{KZ}} = d - \sum_{i<j} \frac{\Omega_{ij}}{k+h^\vee}\, d\log(z_i - z_j).$$

On evaluation modules, monodromy gives

$$q = \exp\left(\frac{\pi i}{k+h^\vee}\right),$$

and the reduced line category is

$$C^{\mathrm{red}}_{\mathrm{line}} \simeq \mathrm{Rep}_q(\mathfrak{g}).$$

The RTT relation must be written as

$$R_{12}(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R_{12}(u-v).$$

The CYBE is

$$[r_{12}(z_{12}), r_{13}(z_{13})] + [r_{12}(z_{12}), r_{23}(z_{23})] + [r_{13}(z_{13}), r_{23}(z_{23})] = 0,$$

and is the Arnold relation applied to the three pairwise logarithmic forms.

12. Virasoro must be treated as an infinite HPL tower

The Virasoro bracket is

$$\{T_\lambda T\} = \partial T + 2\lambda T + \frac{c}{12}\lambda^3.$$

The invariant is

$$\kappa^{\mathrm{Hodge}}_{\mathrm{ch}}(\mathrm{Vir}_c) = \frac{c}{2},$$

not the divided-power coefficient $c/12$.

The transferred minimal A_∞-operations are

$$m^{\min}_n = \sum_{T \in PRT_n} \pm p\, \mu_T(i, \ldots, i; h, \ldots, h),$$

where

$$(Qh + hQ) = 1 - ip.$$

Here $\mu = m_2$ is the binary Virasoro OPE operation, i includes cohomology representatives, p projects to cohomology, and h is the chosen contracting homotopy, usually weight-normalized by L_0^{-1} on positive-weight complement:

$$h|_{A_w} = \frac{1}{w} G_0 \qquad (w > 0),$$

when an antighost G_0 is available.

The first nontrivial class-M obstruction is

$$S_4(\mathrm{Vir}_c) = \frac{10}{c(5c+22)}.$$

The manuscript explicitly says this nonzero class in bar weight 4 forces the class-M completed/pro ambient. main

The line-sector Virasoro R-matrix, as stated in the manuscript, is

$$R(z) = z^{2h} \exp\left(-\frac{c}{4z^2}\right) = z^{2h} \sum_{k\geq 0} \frac{(-c/4)^k}{k!} z^{-2k},$$

with coefficients

$$1, \quad -\frac{c}{4}, \quad \frac{c^2}{32}, \quad -\frac{c^3}{384}, \ldots$$

after factoring z^{2h}. If this remains in the monograph, it needs a Ward-identity derivation from the Virasoro two-point OPE on primary lines and a proof that no odd terms appear. The manuscript gives the formula but not enough proof for publication. main

13. W_3 must be built from the Λ-channel

For W_3, use generators T and W, of weights $2, 3$. Define

$$\Lambda =: TT : -\frac{3}{10}\partial^2 T.$$

The W-W bracket is

$$\{W_\lambda W\} = \frac{c}{360}\lambda^5 + \frac{1}{3}T\lambda^3 + \frac{1}{2}(\partial T)\lambda^2 + \left(\frac{3}{10}\partial^2 T + \frac{32}{22+5c}\Lambda\right)\lambda + \frac{1}{15}\partial^3 T + \frac{16}{22+5c}\partial\Lambda.$$

The first genuinely W_3-specific higher operation is the Λ-mediated transfer. With P_Λ the projection to the quasi-primary line $\mathbb{C}\Lambda$,

$$P_\Lambda(X) = \frac{\langle X, \Lambda\rangle}{\langle \Lambda, \Lambda\rangle}\Lambda,$$

the Λ-channel contribution to the four-point operation is

$$m_{4,\Lambda}(W_1, W_2, W_3, W_4) = \frac{32}{22+5c} \sum_{\text{planar } ab|cd} \pm\, p\, \mu\left(hP_\Lambda\mu(W_a, W_b), hP_\Lambda\mu(W_c, W_d)\right) \cdot K_{ab|cd}(\lambda)$$

where

$$K_{ab|cd}(\lambda) = \mathrm{Res}_{D_{ab|cd}}\left(\omega_{ab} \wedge \omega_{cd}\right)$$

is the corresponding $FM_4(\mathbb{C})$ boundary kernel. This is the missing formula that converts the phrase "$32/(22+5c)$ from the $WW\Lambda$-channel" into an actual operation.

On primaries,

$$\Lambda_0|h\rangle = \left(h^2 - \frac{3h}{5}\right)|h\rangle,$$

so the Λ-channel vanishes at

$$h = 0, \qquad h = \frac{3}{5}.$$

The manuscript's W_3 Hamiltonian calculation records exactly these residues and this Λ_0 -eigenvalue; the monograph should derive the m_4-kernel from them. main

The W_3 Gaudin Hamiltonian must be matrix-valued:

$$H_i^{W_3} = H_i^{TT} + H_i^{TW} + H_i^{WT} + H_i^{WW},$$

with

$$H_i^{W_3} = \sum_{j \neq i} \sum_{k=1}^{5} \frac{c_k^{(W_3)}(V_j, \partial_{z_j})}{z_{ij}^k}.$$

On primary states, the manuscript gives the depth list:

$$\frac{\partial_{z_j}}{z_{ij}}, \qquad \frac{h_j}{z_{ij}^2}, \qquad \frac{w_j}{z_{ij}^3}, \qquad 0, \qquad \frac{c/3}{z_{ij}^5}.$$

The missing theorem is commutativity:

$$[H_i^{W_3}, H_j^{W_3}] = 0.$$

The local PVA Jacobi identity is insufficient; one must construct the W_3 quantum RLL relation. The manuscript itself says this. main

14. The class-M completion must be a Banach estimate

The analytic body is

$$A^{\wedge_\rho} = \left\{ a = \sum_{w \geq 0} a_w : \|a\|_\rho = \sum_{w \geq 0} \|a_w\| \rho^w < \infty \right\}.$$

The operations must satisfy

$$\|m_k(a_1, \ldots, a_k)\|_\rho \leq C_k(\rho) \prod_i \|a_i\|_\rho,$$

and the Maurer–Cartan series must converge:

$$\sum_{k \geq 2} C_k(\rho) < \infty.$$

For Virasoro,

$$\rho_*(c) = \frac{|c|}{6}.$$

For W_3,

$$\rho_*(c) = \frac{|c|}{10}.$$

For principal W_N, the proposed law is

$$\rho_*(W_N) = \frac{|c|}{\beta_N}, \qquad \beta_N = 12(H_N - 1),$$

but unless the harmonic beta-law is proved in the finite-envelope locus, this is a named hypothesis, not a theorem.

15. The DS–Hochschild theorem must be an HPL transfer

For $W_k(\mathfrak{g}, f)$,

$$W_k(\mathfrak{g}, f) = H^0\left(V_k(\mathfrak{g}) \otimes \mathcal{F}_{\mathrm{gh}}, Q_{\mathrm{DS}}\right).$$

Choose a good grading

$$\mathfrak{g} = \bigoplus_{j \in \frac{1}{q}\mathbb{Z}} \mathfrak{g}_j,$$

and a homogeneous basis u_α of $\mathfrak{g}_{>0}$. With ghosts $\psi^\alpha, \psi^*_\alpha$,

$$Q_{\mathrm{DS}} = \mathrm{Res}_{z=0} \quad \sum_\alpha \big(J^{u_\alpha}(z) - \chi(u_\alpha)\big)\psi^*_\alpha(z) - \frac{1}{2}\sum_{\alpha,\beta,\gamma} f^\gamma_{\alpha\beta} : \psi^*_\alpha(z)\psi^*_\beta(z)\psi^\gamma(z) : + \cdots \quad dz.$$

The Hochschild comparison requires an SDR

$$(C^\bullet_{\mathrm{ch}}(V_k, V_k) \otimes \mathcal{F}_{\mathrm{gh}}, Q_{\mathrm{DS}}) \underset{i}{\overset{p}{\rightleftarrows}} C^\bullet_{\mathrm{ch}}(W_k, W_k), \qquad h,$$

with

$$Q_{\mathrm{DS}}h + hQ_{\mathrm{DS}} = 1 - ip.$$

Then braces transfer by

$$\mu^W_n = \sum_T \pm p\, \mu^V_T(i, \ldots, i; h, \ldots, h).$$

The line to print is

$$C^\bullet_{\mathrm{ch}}(W_k, W_k) \simeq H^\bullet_{\mathrm{DS}} C^\bullet_{\mathrm{ch}}(V_k, V_k)$$

as brace E_2^{ch}-algebras, and as $SC_{\mathrm{ch,top}}$-pairs only after h_{oc} and the completed mixed estimates are supplied. The manuscript's proof-obligation section already says the mixed lift uses the same two-coloured cobar homotopy. main

For non-principal f, the good grading may be fractional. The concrete repair is the branched-cover integralization:

$$t = s^q,$$

so that a field of Kazhdan degree j/q becomes integral degree j upstairs. Construct

$$G'_f$$

on the s-cover, prove

$$T_{\mathrm{DS}} = [Q_{\mathrm{DS}}, G'_f]$$

there, check μ_q-invariance, and descend. This is the actual mechanism behind the Bershadsky–Polyakov and hook-type cases.

16. The modular graph complex must be explicit

For a stable graph Γ, define

$$\mathfrak{o}(\Gamma) = \det(E(\Gamma)) \otimes \bigotimes_{v \in V(\Gamma)} \det(H_1(\Gamma_v))^{\otimes -1}.$$

The relative Feynman transform is

$$FT_{\mathrm{rel}}(C) = \prod_{[\Gamma]} {\mathfrak{o}(\Gamma) \otimes \bigotimes_{v \in V(\Gamma)} sC(g(v), \mathrm{Fl}(v))}_{\mathrm{Aut}\,\Gamma}.$$

The differential is

$$D = D_{\mathrm{int}} + D_{\mathrm{sep}} + D_{\mathrm{nsep}},$$

where

$$D_{\mathrm{sep}} = \sum_{e \text{ separating}} \pm \operatorname{contract}_e,$$

and

$$D_{\mathrm{nsep}} = \sum_{e \text{ nonseparating}} \pm \xi_e,$$

with ξ_e the clutching map increasing genus by one.

The identities are

$$D_0^2 = 0, \qquad D_1^2 = 0, \qquad D_0 D_1 + D_1 D_0 = 0,$$

for

$$D_0 = D_{\text{int}} + D_{\text{sep}}, \qquad D_1 = D_{\text{nsep}}.$$

The modular MC element is

$$\Theta = \sum_{g \geq 0} \hbar^g \Theta^{(g)}.$$

The recursion is

$$D_0\Theta^{(0)} + \frac{1}{2}[\Theta^{(0)}, \Theta^{(0)}] = 0,$$

$$D_0\Theta^{(1)} + D_1\Theta^{(0)} + [\Theta^{(0)}, \Theta^{(1)}] = 0,$$

$$D_0\Theta^{(g)} + D_1\Theta^{(g-1)} + \frac{1}{2}\sum_{a+b=g}[\Theta^{(a)}, \Theta^{(b)}] = 0.$$

The obstruction is

$$[o_g] = \quad D_1\Theta^{(g-1)} + \frac{1}{2}\sum_{\substack{a+b=g \\ a,b<g}}[\Theta^{(a)}, \Theta^{(b)}] \quad .$$

This belongs in a theorem, not in synthesis prose.

17. Genus one must use an actual Arakelov kernel

On an elliptic curve

$$E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}),$$

the Arakelov Green function is

$$G_\tau(z) = -\log \quad {\frac{\theta_1(z|\tau)}{\eta(\tau)}}^{2} + \frac{2\pi(\operatorname{Im} z)^2}{\operatorname{Im}\tau}.$$

It satisfies

$$\partial\bar{\partial}G_\tau = \pi i(\delta_0 - \mu_\tau).$$

The chiral propagator is

$$P_\tau(z) = \partial_z G_\tau(z)\, dz,$$

and its failure to be globally exact is the curvature term

$$d_{\text{fib}}^2 = \kappa_{\text{ch}}^{\text{Hodge}}(A)\,\omega_1.$$

The scalar uniform-weight genus tower is

$$F_g(A) = \kappa_{\text{ch}}^{\text{Hodge}}(A)\lambda_g^{FP},$$

with

$$\lambda_g^{FP} = \int_{M_{g,1}} \psi_1^{2g-2}\lambda_g = \frac{2^{2g-1}-1}{2^{2g-1}}\frac{|B_{2g}|}{(2g)!}.$$

The manuscript computes these coefficients and their asymptotics; this should become the scalar genus theorem. main

For multi-weight algebras, the scalar formula is incomplete:

$$F_g(A) = \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}(A)\lambda_g^{FP} + \delta F_g^{\mathrm{cross}}(A).$$

For W_3, the genus-two cross-channel term recorded in the manuscript is

$$\delta F_2^{\mathrm{cross}}(W_3) = \frac{c+204}{16c}.$$

That term is not optional. It is the first place where the tensor-Arakelov curvature is visible.

18. The tensor-Arakelov curvature must replace scalar κ

The scalar equation is only

$$\mathrm{tr}_F\, K(A) = \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}(A)\omega_g.$$

The real object is

$$K(A) \in \mathrm{Sym}^2(F^\vee) \otimes \Omega^2(M_{g,n}).$$

For a basis of fields e_α, write

$$K(A) = \sum_{\alpha,\beta} K_{\alpha\beta}(A)\, e^\alpha e^\beta.$$

Then

$$K_{\alpha\beta}(A) = \sum_{\Gamma\in G_{g,2}^{\alpha,\beta}} \frac{1}{|\operatorname{Aut}\Gamma|}\int_{[M_\Gamma]} \prod_{e\in E(\Gamma)} P_{\alpha_e\beta_e} \prod_{v\in V(\Gamma)} C_v.$$

Uniform-weight theories have

$$K_{\alpha\beta} = \kappa\, \delta_{\alpha\beta}\omega_g.$$

W_3 does not. It has TT, TW, WT, and WW channels. The scalar trace forgets the mixed channels.

19. The anomaly completion must satisfy the curved MC equation

If a curved complex satisfies

$$d^2 = [\Theta, -],$$

then adjoining η must be governed by the curved Maurer–Cartan equation

$$d\eta + \frac{1}{2}[\eta, \eta] + \Theta = 0.$$

Define

$$d_\eta = d + [\eta, -].$$

Then

$$d_\eta^2 = [d\eta + \frac{1}{2}[\eta, \eta] + \Theta, -] = 0.$$

If the manuscript writes simply $d\eta = \Theta$ and $u = \eta^2$, it must specify the bracket convention under which the missing $\frac{1}{2}[\eta, \eta]$ term is absorbed. The secondary anomaly is

$$u = \eta^2$$

only after the algebra structure on the transgression algebra is fixed.

For Virasoro, the anomaly-completed class-M complex should be

$$B_\Theta^{\mathrm{Vir}} = B_{\mathrm{ch}}(\mathrm{Vir}_c)^{\wedge_\rho} \otimes k[\eta, u]/(u - \eta^2),$$

with

$$d\eta + \eta^2 + \Theta_{\mathrm{Vir}} = 0.$$

20. The K3/Borcherds sector needs an operator, not a scalar identity

The scalar identity is

$$Z_{K3\times E}^{\mathrm{BPS}}(Z) = \frac{1}{\Phi_{10}^{\mathrm{un}}(Z)} = \Delta_5(Z)^{-2}.$$

The object-level theorem must construct an operator

$$\mathcal{D}_{K3\times E}$$

in a cyclic chiral Hochschild complex such that

$$\mathrm{Pf}^{\mathrm{prot}}(\mathcal{D}_{K3\times E}) = \Delta_5$$

as a chain-level identity, not merely after taking a scalar character.

The Borcherds product must appear as

$$\Delta_5(Z) = e^{2\pi i(\rho,Z)} \prod_{\alpha\in L_+} \left(1 - e^{2\pi i(\alpha,Z)}\right)^{c(-\alpha^2/2)}.$$

The Hall side must construct

$$\mathrm{CoHA}(K3) \to D_{\mathrm{Hall}}(K3) \to \mathfrak{g}_{\Delta_5},$$

and the chiral side must construct

$$D^b\,\mathrm{Coh}(K3) \xrightarrow{\Phi^{FA}} E_d\text{-FactAlg} \xrightarrow{\mathrm{SpCh}} \mathrm{ChirAlg}_C \xrightarrow{Z^{\mathrm{der}}_{\mathrm{ch}}} SC_{\mathrm{ch,top}}\text{-Alg}\,.$$

The missing theorem is commutativity of the square

$$\begin{tikzcd}
D^b\mathrm{Coh}(K3) \arrow[d] & \mathfrak{g}_{\Delta_5}\text{-Mod} \arrow[d] \\
Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{SpCh}\Phi^{FA}(K3)) & SC_{\mathrm{ch,top}}\text{-Alg.}
\end{tikzcd}$$

Without this square, $1/\Phi_{10}$ is a scalar shadow only.

21. The celestial bridge must be a residue theorem

For a four-dimensional amplitude $A(\omega_i, z_i, \bar{z}_i)$, define the celestial transform

$$\mathcal{M}[A](\Delta_i, z_i, \bar{z}_i) = \prod_i \int_0^\infty \omega_i^{\Delta_i - 1}\, d\omega_i\, A(\omega_i, z_i, \bar{z}_i).$$

Soft operators are residues at special conformal dimensions:

$$S_r(z) = \mathrm{Res}_{\Delta=1-r}\, \mathcal{O}_\Delta(z).$$

The proposed dictionary is

$$m_r \longleftrightarrow S^{\mathrm{soft}}_{r-2}.$$

To make this a theorem, one needs

$$\mathrm{Res}_{\Delta=1-r}\, \mathcal{M}[A_{n+1}] = \sum_{i=1}^n \mathcal{D}_{r,i}\, \mathcal{M}[A_n],$$

and

$$\mathcal{D}_{r,i} = \rho_i\big(\pi_1 D_A|_{(s^{-1}A)^{\otimes r}}\big).$$

For Virasoro,

$$S_2 = \frac{c}{2},$$

$$S_3 \text{ is gauge-trivial},$$

$$S_4 = \frac{10}{c(5c+22)},$$

$$S_5 = -\frac{48}{c^2(5c+22)}.$$

The manuscript records the soft hierarchy and the quartic contact term, but the theorem must be the Mellin residue identity above. main

22. The physical gravity claims must be separated into exact mathematical claims

For $A = \mathrm{Vir}_c$,

$$c = \frac{3\ell}{2G_N}$$

is the Brown–Henneaux dictionary. The rigorous algebraic statement is

$$Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c)^{\wedge_\rho} \in E_3^{\mathrm{top}}\text{-Alg}, \qquad 0 < \rho < \frac{|c|}{6}.$$

The physical reading "pure 3d gravity" is the boundary-CFT holographic interpretation, not a constructed dynamical-metric path integral. The manuscript itself states this scope qualifier. main

The scalar partition function must be defined as

$$Z^{\mathrm{grav}}_{\mathrm{bar}}(\hbar) = \mathrm{Tr}_{B_{\mathrm{ch}}(\mathrm{Vir}_c)^{\wedge_\rho}} \exp\left(\sum_{g\geq 0} \hbar^g \Theta^{(g)}\right).$$

It is not automatically the gravitational path integral. Equality to a BTZ or de Sitter saddle requires separate hypotheses:

$$\mathrm{BHdict}, \quad \mathrm{ModInv}, \quad \mathrm{VacDom}, \quad \mathrm{SadDom}, \quad \mathrm{Borel/Stokes}.$$

The concrete replacement paragraph for the manuscript

The following paragraph is the level at which the monograph should speak.

Let A be a complete filtered E_1-chiral algebra on X. The ordered chiral bar coalgebra is

$$B_{\mathrm{ch}}(A) = T^c(s^{-1}\bar{A}),$$

with coderivation

$$D_A = \sum_{k \geq 1} D_{m_k},$$

where

$$\pi_1 D_A|_{(s^{-1}A)^{\otimes k}} = s^{-1} m_k s^{\otimes k}.$$

For a collision divisor $D_S \subset FM_k(\mathbb{C})$,

$$m_S = \operatorname{Res}_{D_S} \quad \prod_{e \in E(S)} d\log(z_{s(e)} - z_{t(e)}) \otimes \mathrm{OPE}_S \quad ,$$

with

$$\operatorname{Res}_{D_S}(d\log\varepsilon_S \wedge \beta + \alpha) = \beta|_{\varepsilon_S = 0}$$

and orientation

$$\mathrm{or}(FM_k) = (-d\varepsilon_S) \wedge \mathrm{or}(D_S).$$

The identity $D_A^2 = 0$ follows from

$$\partial^2[FM_k(\mathbb{C})] = 0$$

and from the Arnold relations

$$\omega_{ij} \wedge \omega_{jk} + \omega_{jk} \wedge \omega_{ki} + \omega_{ki} \wedge \omega_{ij} = 0.$$

The derived centre is

$$Z^{\mathrm{der}}_{\mathrm{ch}}(A) = \operatorname{Coder} B_{\mathrm{ch}}(A), \qquad d = [D_A, -],$$

with braces

$$f\{g_1, \ldots, g_r\} = \sum_{\text{ordered insertions}} \pm f(1, \ldots, g_1, \ldots, g_r, \ldots, 1).$$

The mixed operations are

$$\nu_k^{(m)} : Z^{\mathrm{der}}_{\mathrm{ch}}(A)^{\otimes k} \otimes A^{\otimes m} \to A.$$

They assemble into a strict $W(SC_{\mathrm{ch,top}})$-algebra precisely after choosing a Drinfeld associator and a two-coloured contracting homotopy

$$h_{\mathrm{oc}} : \Omega((SC_{\mathrm{ch,top}})^!) \to \Omega((SC_{\mathrm{ch,top}})^!)[-1]$$

satisfying

$$dh_{\mathrm{oc}} + h_{\mathrm{oc}} d = 1 - ip.$$

If $T = [Q, G]$ and translations are Q-exact in the named ambient, then Dunn additivity gives

$$Z_{\mathrm{ch}}^{\mathrm{der}}(A) \in E_3^{\mathrm{top}}\text{-Alg}\,.$$

For class M, all these identities are in $A^{\wedge_\rho}$, with

$$\|m_k\|_\rho \le C_k(\rho), \qquad \sum_k C_k(\rho) < \infty.$$

This is the skeleton with bones, tissue, and muscle.

Sources

you started in the right direction, but you have left out IMMENSE detail. fill in all the gaps without repeating anything you stated earlier.

Thought for 8m 19s >

According to a document from 2026-06-05, the parts that still need real mathematical body are the **local exact sector**, the **line algebra/Kuranishi sector**, the **Baxter–Rees/shifted RTT sector**, the **KZB wild-boundary sector**, the **anomaly-completed Steinberg sector**, the **finite-orbifold descent sector**, and the **operator-level K3/Borcherds sector**. The manuscript's own notation already demands stage licensing $P \to C \to S \to Z \to A$, conditional tags, and the refusal of shadow/object collapse; the missing material is therefore not another universal slogan but the actual formulas below. main

I will not restate the definitions already given earlier: the basic bar coalgebra, the basic $SC_{\mathrm{ch,top}}$ colours, the elementary FM residue convention, the elementary Virasoro/W_3 brackets, or the genus-expanded MC recursion. The following are the missing lower strata.

1. Boundary-linear local sector: Kuranishi algebra, not metaphor

Let a boundary germ be given by a formal map

$$F : E \longrightarrow O, \qquad F(p) = 0.$$

Choose splittings

$$E = T \oplus I, \qquad O = B \oplus C,$$

where

$$T = \ker(dF_p), \qquad B = \mathrm{im}(dF_p), \qquad C = \mathrm{coker}(dF_p),$$

and

$$A := dF_p|_I : I \xrightarrow{\sim} B.$$

Write Taylor components

$$F = \sum_{n\geq 1} \frac{1}{n!} F_n, \qquad F_n : E^{\otimes n} \to O,$$

and decompose

$$F_n = F_n^B + F_n^C.$$

The massive field $i(t) \in I[[T^\vee]]$ is determined recursively by

$$F^B(t + i(t)) = 0,$$

with

$$i_2(t_1, t_2) = -A^{-1} F_2^B(t_1, t_2),$$

and

$$i_3(t_1, t_2, t_3) = -A^{-1} \left[F_3^B(t_1, t_2, t_3) + \sum_{\text{cyc}} F_2^B\big(t_1, i_2(t_2, t_3)\big) \right].$$

The reduced Kuranishi map is

$$\kappa : T \longrightarrow C, \qquad \kappa(t) = F^C(t + i(t)).$$

Thus

$$\kappa_2(t_1, t_2) = F_2^C(t_1, t_2),$$

and

$$\kappa_3(t_1, t_2, t_3) = F_3^C(t_1, t_2, t_3) - \sum_{\text{cyc}} F_2^C\big(t_1, A^{-1} F_2^B(t_2, t_3)\big).$$

This is the field-theoretic "integrating out a massive mode" formula. The cubic effective vertex is not the raw F_3^C; it is corrected by the tree-level exchange

$$T \otimes T \xrightarrow{F_2^B} B \xrightarrow{A^{-1}} I \xrightarrow{F_2^C} C.$$

The minimal pointed line algebra is

$$K_\kappa = \mathrm{Sym}(C) \otimes \Lambda(T),$$

with operations

$$m_2^\kappa(\lambda t_1, \lambda t_2) = \lambda t_1 \wedge \lambda t_2 + \frac{1}{2} \kappa_2(t_1, t_2),$$

$$m_3^\kappa(\lambda t_1, \lambda t_2, \lambda t_3) = \frac{1}{6}\kappa_3(t_1, t_2, t_3),$$

and, in general,

$$m_n^\kappa(\lambda t_1, \dots, \lambda t_n) = \frac{1}{n!}\kappa_n(t_1, \dots, t_n) \qquad (n \geq 2),$$

up to the exterior product part in arity 2. This is exactly the local line algebra hidden beneath the boundary-linear sector. The manuscript identifies this as the minimal pointed line algebra and records the first effective couplings in precisely this Kuranishi form. main

The effective boundary-linear potential is

$$W_{\text{eff}}(u, \gamma) = \langle \gamma, \kappa(u) \rangle, \qquad \gamma \in C^\vee.$$

The completed local bulk algebra is

$$\mathcal{O}\Big(d\,\mathrm{Crit}(W_{\text{eff}})\Big) = \Big(\mathrm{Sym}(T^\vee \oplus C^\vee) \otimes \Lambda(T \oplus C), d_{W_{\text{eff}}}\Big),$$

where

$$d_{W_{\text{eff}}} = \sum_i \frac{\partial W_{\text{eff}}}{\partial u_i}\frac{\partial}{\partial \xi_i} + \sum_a \frac{\partial W_{\text{eff}}}{\partial \gamma_a}\frac{\partial}{\partial \chi_a}.$$

The missing theorem should read

$$HH^\bullet(K_{F,p}) \simeq HH^\bullet(K_\kappa) \simeq HH^\bullet(B_\kappa) \simeq \mathcal{O}\Big(d\,\mathrm{Crit}(W_{\text{eff}})\Big).$$

This is the local bulk-from-lines theorem; the bulk is Hochschild of the line algebra, not the line algebra itself. The manuscript states this local bulk/line theorem and the stable invisibility of hyperbolic blocks, but it needs to be pushed into the main construction rather than left as a side sector. main

The boundary-linear Mather–Yau statement is:

$$K_\kappa \simeq K_{\kappa'} \quad \text{as polarized } A_\infty\text{-algebras}$$

if and only if the reduced microlocal formal germs

$$\kappa : T \to C, \qquad \kappa' : T' \to C'$$

are formally equivalent after stabilization by tangent-normal and pure-normal hyperbolic blocks. In coordinates, a polarized A_∞-isomorphism must preserve the exterior generators up to

$$T \overset{\sim}{\quad} T',$$

the symmetric obstruction variables up to

$$C \overset{\sim}{\quad} C',$$

and all higher Taylor tensors

$$[\kappa_n] \in \mathrm{Sym}^n(T^\vee) \otimes C$$

up to the usual right-left action. The manuscript's "fortified local triangle" already contains this chain

$$(F, p) \rightsquigarrow J_{F,p} \rightsquigarrow K^{\mathrm{ex}}_{F,p} \simeq K_{\kappa_F} \rightsquigarrow HH^\bullet(K_{\kappa_F}) \simeq \mathcal{O}(d\,\mathrm{Crit}\,W_{\mathrm{eff},F}),$$

and it must become the local model for every boundary condition. main

2. Bordered configuration spaces and the diagonal bicomodule

The open-sector geometry should not be described only by ordinary ordered intervals. The missing object is the bordered Fulton–MacPherson compactification

$$\mathrm{Conf}_{p,q}(\mathbb{H})$$

of p interior points and q ordered boundary points in the upper half-plane. A boundary stratum is indexed by a planted forest with two root types:

$$\mathcal{F} = (V_{\mathrm{int}}, V_\partial, E; \rho_{\mathrm{bulk}}, \rho_\partial),$$

where vertices over ρ_{bulk} record interior collisions and vertices over ρ_∂ record boundary collisions. The codimension-one strata are of three types:

$$\text{interior–interior: } z_i \to z_j,$$

$$\text{boundary–boundary: } x_a \to x_b,$$

$$\text{interior–boundary: } \mathrm{Im}\, z_i \to 0, \quad \mathrm{Re}\, z_i \to x_a.$$

The diagonal bicomodule is

$$\Delta_A \in {}_A\,\mathrm{Bimod}_A,$$

with underlying object A, left action from negative boundary approach and right action from positive boundary approach. Its geometric chain model is

$$D^{p,q}_A = C^{\log}_\bullet\big(\mathrm{Conf}_{p,q}(\mathbb{H})\big) \otimes A^{\otimes p} \otimes A^{\otimes q}.$$

The differential decomposes as

$$d_D = d_{\mathrm{int}} + d_\partial + d_{\mathrm{mix}},$$

where

$$d_{\mathrm{int}} = \sum_{S \subset \{1,\ldots,p\}} \mathrm{Res}_{D^{\mathrm{int}}_S} \otimes m_S,$$

$$d_\partial = \sum_{I \subset \{1,\dots,q\}} \mathrm{Res}_{D_I^\partial} \otimes \mu_I,$$

and

$$d_{\mathrm{mix}} = \sum_{S,I} \mathrm{Res}_{D_{S,I}^{\mathrm{mix}}} \otimes \nu_{S,I}.$$

Here $\nu_{S,I}$ is the bulk-to-boundary operation obtained by taking interior fields in S to a boundary cluster I. This is the missing chain model for the statement "the diagonal is the universal boundary condition."

The annular closure is obtained by replacing the interval with S^1:

$$D_A^{\mathrm{ann}} = \bigoplus_n C_\bullet^{\log}\big(\mathrm{Conf}_n(S^1 \times \mathbb{R}_{\geq 0})\big) \otimes_{\mathbb{Z}/n} A^{\otimes n}.$$

The cyclic coinvariants are not harmless: the cyclic rotation contributes the sign

$$[s^{-1}a_1|\cdots|s^{-1}a_n] \mapsto (-1)^{(\bar{a}_n)(\bar{a}_1+\cdots+\bar{a}_{n-1})}[s^{-1}a_n|s^{-1}a_1|\cdots|s^{-1}a_{n-1}],$$

with $\bar{a}_i = |a_i| - 1$. The annular monodromy is where the genus-one R-matrix and the trace pairing first appear.

3. Punctured disk versus circle: the E_2-to-E_1 loss

The punctured-disk chain model is

$$B_{D^*}(A) = \bigoplus_n C_\bullet\big(\mathrm{Conf}_n(D^*)\big) \otimes_{\Sigma_n} A^{\otimes n}.$$

There is a radial projection

$$\rho : \mathrm{Conf}_n(D^*) \to \mathrm{Conf}_n(S^1)$$

which gives a quasi-isomorphism on the "angular" subcomplex, but it kills the radial collision class. The missing distinction is:

$$H_\bullet(\mathrm{Conf}_n(D^*)) \neq H_\bullet(\mathrm{Conf}_n(S^1))$$

as operadic modules, even when the chosen bar chain model is quasi-isomorphic after forgetting the E_2-rotation. The kernel is generated by the winding class

$$\theta_i = d \arg z_i$$

and the pairwise logarithmic classes

$$d \log(z_i - z_j).$$

The E_2-to-E_1 gap is the loss of the BV operator

$$\Delta_{\mathrm{rot}} = \sum_i \iota_{\partial_{\arg z_i}},$$

which survives on D^* but becomes ordinary cyclic rotation on S^1. Thus the circle bar computes the annular trace; the punctured disk computes the chiral centre with rotation. Those are not the same chain model.

4. Type-A Baxter–Rees compactification

The type-A line sector must be written in RTT coordinates. Let

$$T(u) = \sum_{i,j=1}^{N} E_{ij} \otimes T_{ij}(u), \qquad T_{ij}(u) = \delta_{ij} + \sum_{r \geq 1} T_{ij}^{(r)} u^{-r}.$$

With

$$R(u) = 1 - \frac{\hbar P}{u},$$

the RTT relation is

$$R(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R(u-v).$$

Equivalently,

$$(u-v)[T_{ij}(u), T_{kl}(v)] = \hbar\big(T_{kj}(u)T_{il}(v) - T_{kj}(v)T_{il}(u)\big).$$

Introduce the weight filtration

$$\deg T_{ij}^{(r)} = r,$$

and the Rees algebra

$$\mathcal{R}_\beta Y_\hbar(\mathfrak{gl}_N) = \bigoplus_{d \geq 0} \beta^d F_d Y_\hbar(\mathfrak{gl}_N).$$

The Baxter–Rees compactification is the formal family

$$\mathfrak{Y}_{\mathrm{Bax}} = \operatorname{Spf} \mathcal{R}_\beta Y_\hbar(\mathfrak{gl}_N) \longrightarrow \operatorname{Spf} \mathbb{C}[[\beta]],$$

whose generic fibre is the RTT Yangian and whose special fibre records the asymptotic/prefundamental boundary category.

The continuation theorem must be weightwise. For every conformal-weight window w, there should be a finite integer $D(w)$ such that

$$R_{V,W}^{(\leq w)}(u,\beta) = \sum_{d=0}^{D(w)} \beta^d R_d^{(\leq w)}(u)$$

satisfies RTT modulo $F_{>w}$. The same must hold for the associator:

$$\Theta_\Phi^{(\le w)}(\beta) = \sum_{d=0}^{D'(w)} \beta^d \Theta_d^{(\le w)}.$$

The boundary tangent class is the first-order deformation

$$\dot{\Theta} = \frac{d}{d\beta}\Big|_{\beta=0} \Theta_\Phi(\beta),$$

and it must satisfy

$$d_{\Theta_0}\dot{\Theta} = 0$$

in the RTT deformation complex. The obstruction to extending it is

$$\frac{1}{2}[\dot{\Theta}, \dot{\Theta}] + d_{\Theta_0}\Theta_2 \in H^2_{\mathrm{RTT}}.$$

This is what the "boundary tensor geometry" must mean: a concrete obstruction tower in the RTT deformation complex, not a phrase. The manuscript itself separates the weaker RTT/factorisation definition from the stronger modular-MC definition of a chiral Yangian and says their chain-level equivalence is still conditional even in type A.

main

5. Shifted RTT and the rank-one Kleinian test

For a coweight shift

$$\mu = (\mu_1, \dots, \mu_N),$$

the shifted RTT algebra is not obtained by an informal truncation. It is the quotient of the completed RTT algebra by the pairing annihilator

I_\mu = \{x\in Y_{\hbar}^{RTT}: \langle x,y\rangle=0 \ \forall\,y\in Y_{\hbar}^{RTT,\vee}_{\ge\mu}\}.

The condition that this quotient define a line algebra is the coideal condition

$$\Delta(I_\mu) \subset I_\mu \otimes Y_\hbar + Y_\hbar \otimes I_\mu.$$

The local shifted generators should be written as

$$T_{ij}(u) = u^{-\mu_i+\mu_j}\left(\delta_{ij} + \sum_{r\ge 1} T_{ij}^{(r)} u^{-r}\right)$$

after choosing a chamber. Boundary quotient data enter through the quantum determinant

$$\operatorname{qdet} T(u) = \sum_{\sigma \in S_N} (-1)^\sigma T_{\sigma(1),1}(u) T_{\sigma(2),2}(u-\hbar) \cdots T_{\sigma(N),N}(u-(N-1)\hbar),$$

which is fixed to a polynomial $P_\mu(u)$:

$$\operatorname{qdet} T(u) = P_\mu(u).$$

In rank one, the naive Cartan quotient loses the coideal property. The correct A_1 boundary quotient is determinantal/Casimir:

$$\mathcal{A}_m^\hbar = \mathbb{C}\langle x, y, z\rangle \Big/ \left([z,x] - \hbar x,\ [z,y] + \hbar y,\ yx - \prod_{a=0}^{m-1} (z - a\hbar),\ xy - \prod_{a=1}^{m} (z + a\hbar) \right).$$

Its associated graded is

$$\operatorname{gr} \mathcal{A}_m^\hbar \simeq \mathbb{C}[x,y,z]/(xy - z^m),$$

the Kleinian A_{m-1} surface. This is the rank-one test: if the proposed shifted RTT quotient does not degenerate to the Kleinian coordinate ring and does not satisfy the coideal condition, the shifted line algebra is not correct. The manuscript explicitly isolates the rank-one Kleinian test and the pairing-annihilator descent as separate from completion issues. main

6. Strong versus weak chiral Yangian

The weaker object is

$$(\mathcal{A}, R(z), \otimes_z),$$

where

$$R_{12}(z-w) R_{13}(z) R_{23}(w) = R_{23}(w) R_{13}(z) R_{12}(z-w).$$

The stronger object is

$$\mathbb{Y}_{\mathrm{ch}} = (\mathcal{A}, \Delta_z, R(z), S(z), \epsilon, \{m_k\}_{k\geq 2}, \Theta_{\mathrm{mod}}),$$

with

$$d\Theta_{\mathrm{mod}} + \frac{1}{2}[\Theta_{\mathrm{mod}}, \Theta_{\mathrm{mod}}] = 0$$

in the completed modular deformation dg Lie algebra. The missing theorem is a map

$$\mathrm{RTTDef}(\mathcal{A}, R) \longrightarrow \mathrm{MC}_{\mathrm{mod}}(\mathcal{A})$$

whose tangent map sends the RTT infinitesimal

$$\dot{R}(z)$$

to the modular first Taylor coefficient

$$\Theta_1.$$

The obstruction is

$$\mathrm{ob}_2(R) = \left[d_{\Theta_0}\Theta_2 + \frac{1}{2}[\Theta_1, \Theta_1] \right] \in H^2_{\mathrm{mod}}(\mathcal{A}).$$

Thus “monodromy equals R-matrix” is a theorem only when this obstruction vanishes and the KZ/KZB monodromy functor lands in the strong modular-MC object. The manuscript distinguishes these two chiral-Yangian notions and says the HT landscape produces the stronger one only when the modular MC tower is actually built. main

7. Super-Yangian signs and the missing Berezinian bridge

For a superspace $V = V_{\bar{0}} \oplus V_{\bar{1}}$, let

$$P_s(v \otimes w) = (-1)^{|v||w|} w \otimes v.$$

The rational super R-matrix is

$$R(u) = 1 - \frac{\hbar P_s}{u}.$$

The RTT relation is

$$R(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R(u-v),$$

where matrix multiplication is graded:

$$(E_{ij} \otimes a)(E_{kl} \otimes b) = (-1)^{(|a|+|i|+|j|)(|k|+|l|)} \delta_{jk} E_{il} \otimes ab.$$

The super-CYBE is

$$[r_{12}(z_{12}), r_{13}(z_{13})]_s + [r_{12}(z_{12}), r_{23}(z_{23})]_s + [r_{13}(z_{13}), r_{23}(z_{23})]_s = 0.$$

The supertrace pairing is

$$\mathrm{str}(X) = \mathrm{tr}(X|_{V_{\bar{0}}}) - \mathrm{tr}(X|_{V_{\bar{1}}}),$$

whereas the Berezinian pairing is governed infinitesimally by

$$d \log \mathrm{Ber}(1 + \epsilon X) = \mathrm{str}(X).$$

The missing bridge is a chain-level pairing

$$\langle -, - \rangle_{\mathrm{can}} : A \otimes A^! \to \mathbb{C}$$

such that

$$\mathrm{Tr}_{A/A_{\mathrm{nil}}} \langle a, a^! \rangle_{\mathrm{can}} = \mathrm{str}(aa^!)$$

and

$$\det_{\mathrm{Ber}}\big(\langle e_i, e_j^! \rangle_{\mathrm{can}}\big) = \mathrm{Ber}(A).$$

The compatibility condition is

$$\langle m_k(a_1, \ldots, a_k), b \rangle = (-1)^{\epsilon} \langle a_1 \otimes \cdots \otimes a_k, \Delta_k b \rangle,$$

with

$$\epsilon = |b|(\bar{a}_1 + \cdots + \bar{a}_k) + \sum_{i<j} |a_i||a_j|.$$

Without this pairing, super-complementarity is only two parallel scalar identities, not one canonical duality.

8. Half-space BV: image-charge propagator and reflected weights

On the half-space

$$\mathbb{H} = \{t \geq 0\}$$

with holomorphic coordinate z, the doubled propagator must be

$$P_{\mathbb{H}}(x, y) = P_{\mathbb{R}\times\mathbb{C}}(x, y) - \sigma\, P_{\mathbb{R}\times\mathbb{C}}(x, \bar{y}),$$

where σ is the boundary reflection acting on fields. For a field ϕ of reflected weight w_ϕ,

$$\sigma(\phi) = (-1)^{w_\phi} \phi.$$

The boundary OPE coefficient is therefore

$$\mathrm{OPE}_\partial(\phi_i, \phi_j) = \mathrm{OPE}_{\mathrm{bulk}}(\phi_i, \phi_j) - (-1)^{w_j}\, \mathrm{OPE}_{\mathrm{bulk}}(\phi_i, \sigma\phi_j).$$

The reflected-weight identity to prove inductively is

$$w_{\mu\circ\nu} \equiv w_\mu + w_\nu + \operatorname{codim} D_{\mu,\nu} \pmod 2.$$

The first non-affine obstruction is the quadratic reflected term

$$\mathcal{O}_2 = \mathrm{Res}_{D_{12}} \left[P(x_1, x_2) - \sigma P(x_1, \bar{x}_2)\right] \otimes \pi_2,$$

and the cubic obstruction is

$$\mathcal{O}_3 = \sum_{\mathrm{cyc}} \mathrm{Res}_{D_{ij}D_{jk}}\, P_{\mathbb{H}}(x_i, x_j) P_{\mathbb{H}}(x_j, x_k) \otimes [\pi_2, \pi_2].$$

The doubling theorem must say

$$\mathcal{O}_3 = 0$$

by the reflected Arnold relation

$$\omega_{12}^{\mathbb{H}} \wedge \omega_{23}^{\mathbb{H}} + \omega_{23}^{\mathbb{H}} \wedge \omega_{31}^{\mathbb{H}} + \omega_{31}^{\mathbb{H}} \wedge \omega_{12}^{\mathbb{H}} = 0.$$

This is the actual half-space mechanism behind "boundary BV descent." The manuscript's affine half-space theorem and its transport to Virasoro and W_3 depend on precisely this reflected-weight/doubling package. main

9. Irregular KZB: formal type and Stokes cocycle

Near a boundary divisor $q = 0$ of $M_{g,n}$, the KZB connection must be written in the form

$$\nabla = d - \left(\frac{A_m}{q^m} + \frac{A_{m-1}}{q^{m-1}} + \cdots + \frac{A_1}{q} + A_0\right) dq - \sum_{\alpha} B_\alpha(q)\, dt_\alpha.$$

The formal type is

$$\Theta_\partial(q) = \sum_{r=1}^{m-1} \frac{A_{r+1}}{r q^r} + A_1 \log q.$$

The formal gauge form is

$$\nabla \simeq d - d\Theta_\partial - \Lambda \frac{dq}{q}$$

after a formal transformation

$$G(q) \in \mathrm{Aut}(V)[[q^{1/N}]].$$

For each Stokes ray ℓ, choose a sector S_ℓ and a fundamental solution

$$Y_\ell(q) = H_\ell(q) e^{\Theta_\partial(q)} q^\Lambda.$$

The Stokes matrices are

$$S_{\ell,\ell'} = Y_\ell^{-1} Y_{\ell'}.$$

The clutching compatibility required for modular operad composition is

$$\Theta_{\partial,\Gamma_1 \# \Gamma_2} = \Theta_{\partial,\Gamma_1} \oplus \Theta_{\partial,\Gamma_2} \oplus \Theta_{\mathrm{node}},$$

and

$$S_{\ell,\ell'}^{\Gamma_1 \# \Gamma_2} = S_{\ell,\ell'}^{\Gamma_1} S_{\ell,\ell'}^{\Gamma_2} S_{\ell,\ell'}^{\mathrm{node}},$$

with order determined by the tangential basepoint convention. Associativity of modular composition at generic nonrational level is therefore equivalent to the Stokes cocycle identity

$$S_{\ell_1,\ell_2} S_{\ell_2,\ell_3} S_{\ell_3,\ell_1} = 1.$$

The manuscript explicitly distinguishes this irregular-KZB composition theorem from curved-Dunn H^2-vanishing: KZB supplies boundary composition only after formal type and mixed Stokes data are supplied; it does not prove the curved-Dunn obstruction vanishing by itself. main

The covered level loci should be written exactly as:

$$k \in \mathbb{Z}_{\geq 0} \quad \Rightarrow \quad \text{KZ pentagon} + \text{Kazhdan–Lusztig regularity},$$
$$k \in \mathbb{C} \setminus \mathbb{Q} \quad \Rightarrow \quad \text{irregular KZB formal type} + \text{wild groupoid},$$
$$k = -h^{\vee} \quad \Rightarrow \quad \text{Feigin–Frenkel centre}.$$

Rational nonintegral noncritical levels remain outside unless an admissible-level comparison theorem is proved. The manuscript states precisely this separation.

main

10. Genus-two handle primitive

At $g = 2$, the relevant ribbon graphs are the theta graph and the two-loop bouquet:

$$\Gamma_{\flat}^{(2)} = \bullet\!\equiv\!\bullet, \qquad \Gamma_{\sharp}^{(2)} = \bullet \text{ with two loops}.$$

Let

$$\sigma_2$$

be a Katz–Oda primitive of

$$\omega_2 = c_1(\lambda_1|_{M_{2,0}})$$

in the chosen relative curved complex. The chain-level handle-attachment primitive is

$$P_2 = \frac{1}{12}\langle \sigma_2, \mathrm{ev}_{\Gamma_{\flat}^{(2)}} \rangle + \frac{1}{8}\langle \sigma_2, \mathrm{ev}_{\Gamma_{\sharp}^{(2)}} \rangle.$$

The desired identity is

$$d_{\Theta^{(0)}} P_2 = D_1 \Theta^{(1)} + \frac{1}{2}[\Theta^{(1)}, \Theta^{(1)}]$$

in the genus-two associated graded. The coefficients $1/12$ and $1/8$ are not arbitrary: they are the inverse automorphism orders of the theta and bouquet graph contributions after orientation normalization. The manuscript explicitly presents P_2 in this form and warns that its existence is a genus-two bootstrap hypothesis, not a consequence of ordinary Gauss–Manin flatness. main

The higher-genus analogue should be

$$P_g = \sum_{\Gamma \in G_{g,0}^{\mathrm{conn}}} \frac{1}{|\operatorname{Aut}\Gamma|} \langle \sigma_g, \mathrm{ev}_\Gamma \rangle,$$

with

$$d_{\Theta^{(0)}} P_g = D_1 \Theta^{(g-1)} + \frac{1}{2} \sum_{\substack{a+b=g \\ a,b<g}} [\Theta^{(a)}, \Theta^{(b)}].$$

That is the concrete coboundary witness the modular bootstrap must construct.

11. Anomaly-completed Steinberg and genus Clifford algebra

Let B be the bar dg algebra and let

$$\Theta \in Z^2(B) \cap Z(B)$$

be the anomaly. The transgression algebra is not always a polynomial extension. In the noncommutative case it is the Ore extension

$$B_\Theta = B\langle \eta \rangle \Big/ \left(\eta b - (-1)^{|b|} b \eta - \iota_\Theta(b) \right),$$

with

$$|\eta| = 1, \qquad d\eta = \Theta.$$

The secondary anomaly is

$$u = \eta^2.$$

For genus g, the genus-Clifford completion is

$$\mathcal{G}_g(B_\Theta) = B_\Theta \langle \alpha_1, \beta_1, \ldots, \alpha_g, \beta_g \rangle \Big/ \mathcal{R}_g,$$

where

$$\alpha_i \alpha_j + \alpha_j \alpha_i = 0,$$

$$\beta_i \beta_j + \beta_j \beta_i = 0,$$

$$\alpha_i \beta_j + \beta_j \alpha_i = \delta_{ij} u.$$

If u is inverted,

$$\mathcal{G}_g(B_\Theta)[u^{-1}] \simeq \mathrm{Mat}_{2^g}(B_\Theta[u^{-1}]).$$

If $u = 0$,

$$\mathcal{G}_g(B_\Theta)/(u) \simeq (B_\Theta/(u)) \otimes \Lambda(\alpha_1, \beta_1, \ldots, \alpha_g, \beta_g).$$

Thus the topological regime is Morita-trivial after inverting u, whereas the strict gravitational regime is exterior. The manuscript gives exactly this dichotomy for the nonabelian $SU(2)$ anomalous Steinberg, including the boundary algebra $V_k(\mathfrak{sl}_2)$, dual level $-k-4$, modular characteristic $3(k+2)/4$, anomaly class $\Theta = \kappa\omega_1$, and genus-Clifford completion. main

For $SU(2)$, the geometric source is

$$H^3(SU(2);\mathbb{Z}) \cong \mathbb{Z},$$

with generator c_3. Level k selects

$$k\, c_3,$$

and anomaly completion is the cochain-level string trivialization. The transgression generator η is the witness

$$d\eta = \Theta,$$

and $u = \eta^2$ records the quadratic class of the chosen trivialization. The manuscript explicitly interprets this as the nonabelian gerbe/string-structure form of anomaly completion. main

12. Original-complex class M: Kac–Shapovalov estimates

The original-complex question for Virasoro must be phrased using the Kac–Shapovalov form. On a Verma module $M(c,h)$, let

$$L_{-\lambda}|h\rangle = L_{-\lambda_1}\cdots L_{-\lambda_\ell}|h\rangle, \qquad |\lambda| = n.$$

The Gram matrix at level n is

$$G_n(c,h)_{\lambda,\mu} = \langle h|L_{\mu_\ell}\cdots L_{\mu_1}L_{-\lambda_1}\cdots L_{-\lambda_\ell}|h\rangle.$$

The determinant is

$$\det G_n(c,h) = C_n \prod_{\substack{r,s\geq 1\\ rs\leq n}} \big(h - h_{r,s}(c)\big)^{p(n-rs)}.$$

The original direct-sum norm is meaningful only away from the Kac walls

$$h = h_{r,s}(c).$$

For a transferred operation m_k, the norm estimate must be of the form

$$\|m_k\|_{\mathrm{KS},n} \leq C^k \max_{|\lambda|\leq n} \|G_{|\lambda|}(c,h)^{-1}\| \cdot P_k(n,c,h).$$

The direct-sum finite-propagation condition is

$$\forall n\ \exists K(n) \quad \text{such that} \quad \pi_{\le n} m_k|_{A_{\le n}^{\otimes k}} = 0 \quad (k > K(n)).$$

For generic Virasoro this fails because the quartic obstruction survives, and higher transferred operations continue to feed lower finite weights through inverse Shapovalov propagators. Thus the original-complex tempered locus is not the same as the generic central-charge locus; it is a locus where the inverse Gram growth is dominated by the shadow-tower decay.

The manuscript has separate chapters for the weight-four direct-sum obstruction, the original-complex dichotomy, and the Kac–Shapovalov norm; these should be fused into a single analytic theorem:

$$\sum_{k\ge 2} \frac{\|m_k\|_{\mathrm{KS},n}}{k!} < \infty \quad \Longleftrightarrow \quad (c,h) \in \mathcal{T}_{\mathrm{orig}}.$$

Outside $\mathcal{T}_{\mathrm{orig}}$, the theorem must live in the completed ambient.

13. Logarithmic triplet $W(p)$: the actual amplitude estimate

The triplet algebra $W(p)$ contains logarithmic sectors with Jordan L_0-blocks. The obstruction is not simply infinite Zhu or finite Zhu. The dangerous amplitudes are boundary-changing correlators

$$\langle \phi_{0,1}(z_1) \cdots \phi_{0,1}(z_r) \rangle ,$$

where $\phi_{0,1}$ is the logarithmic boundary field. The needed estimate is

$$|\langle \phi_{0,1}(z_1) \cdots \phi_{0,1}(z_r) \rangle| \le C^r (r!)^{1-\epsilon} \prod_{i<j} |z_i - z_j|^{-N_{ij}} \left(1 + \sum_{i<j} |\log|z_i - z_j||\right)^M$$

for some $\epsilon > 0$, uniformly on the regular logarithmic sector after fixing configuration-space cutoffs.

Equivalently, the shadow coefficients must satisfy

$$\limsup_{r\to\infty} \left(\frac{|S_r(W(p))|}{r!}\right)^{1/r} = 0.$$

A finite-dimensional Zhu algebra does not imply this. The manuscript explicitly says logarithmic amplitudes can have unbounded Massey products despite finite Zhu and that the Zhu-bounded-Massey route does not close $W(p)$. main

The correct decomposition is

$$S_r(W(p)) = S_r^{\mathrm{reg}} + S_r^{\mathrm{nil}} + S_r^{\log},$$

where S_r^{reg} is ordinary semisimple propagation, S_r^{nil} comes from nilpotent Jordan insertions, and $S_r^{\log}$ carries explicit powers of $\log z$. The missing theorem is

$$|S_r^{\log}| \le C^r (r!)^{1-\epsilon}.$$

Without this, $W(p)$ remains frontier, not theorem.

14. Irrational cosets: VSKR + BGG as the real criterion

For a coset

$$A = \mathrm{Com}(B, C),$$

the Zhu algebra may be infinite and still not obstruct tempering. The correct criterion is not Zhu-finiteness but a Verma-shadow/Kazhdan-ramification plus BGG resolution estimate.

For the parafermion coset

$$K(\mathfrak{sl}_2, k) = \mathrm{Com}(H, V_k(\mathfrak{sl}_2)),$$

one should choose a BGG resolution

$$\cdots \to \bigoplus_{\ell(w)=2} M(w\cdot\lambda) \to \bigoplus_{\ell(w)=1} M(w\cdot\lambda) \to M(\lambda) \to L(\lambda) \to 0.$$

The coset shadow coefficient is the alternating sum

$$S_r^{\mathrm{coset}} = \sum_{w\in W_{\mathrm{aff}}} (-1)^{\ell(w)} S_r(M(w\cdot\lambda))\, q^{\Delta(w\cdot\lambda)-\Delta(\lambda)}.$$

Tempering follows if the ramification exponents satisfy

$$\mathrm{Re}\big(\Delta(w\cdot\lambda) - \Delta(\lambda)\big) \ge a\,\ell(w)^2 - b$$

for some $a > 0$, because then the BGG alternating sum is dominated by Gaussian decay in affine length. The manuscript notes that irrational cosets enter through the refined VSKR+BGG criterion and that this is strictly weaker than C_2-cofiniteness. main

15. Monster and Schellekens finite-orbifold descent

For a finite group G acting on a holomorphic VOA V, the orbifold descent datum is not merely

$$V^G.$$

It is the homotopy fixed-point object

$$V^{hG} = \mathrm{Tot}\left[V \rightrightarrows \mathrm{Map}(G,V)\triplearrows \mathrm{Map}(G^2,V)\cdots\right].$$

The obstruction is a Dijkgraaf–Witten class

$$[\alpha_{\mathrm{orb}}] \in H^3(BG;U(1)).$$

A chain-level E_3-descent requires

$$[\alpha_{\mathrm{orb}}] = 0.$$

If $\alpha_{\mathrm{orb}} = d\beta$, then the corrected associator is

$$\Phi^{\beta}_{g,h,k} = \beta(g,h)\beta(gh,k)\beta(h,k)^{-1}\beta(g,hk)^{-1}\Phi_{g,h,k}.$$

For the Leech $\mathbb{Z}/2$ Monster construction, the local sign condition is

$$\alpha_{\mathrm{orb}}(\sigma,\sigma,\sigma) = +1.$$

For Schellekens $c = 24$, the descent stratification is

$$71 = 24_{\mathrm{Niemeier}} + 1_{V_1=0} + 46_{\mathrm{nonlat}}.$$

Each cyclic orbifold row needs:

$$\text{VOA presentation} + \text{level matching} + [\alpha_{\mathrm{orb}}] = 0 + \text{finite-orbifold BV descent}.$$

The manuscript explicitly says the Monster case requires the Dijkgraaf–Witten anomaly trivialisation separately from the FLM boundary identification, and Schellekens rows require a three-stratum anomaly/BV descent criterion. main

16. Six-dimensional holomorphic Chern–Simons anomaly

For a Calabi–Yau threefold Y, fields are

$$A \in \Omega^{0,\bullet}(Y,\mathfrak{g})[1],$$

with action

$$S(A) = \int_Y \Omega_Y \wedge \left(\frac{1}{2}\langle A,\bar{\partial}A\rangle + \frac{1}{6}\langle A,[A,A]\rangle\right).$$

The one-loop BV anomaly is a local functional of degree 1:

$$\mathcal{A}^{(1)} = \int_Y \Omega_Y \wedge \mathrm{Tr}_{\mathfrak{g}}\big(A\,\partial A\,\partial A\,\partial A\big),$$

more invariantly a class in

$$H^1_{\mathrm{loc}}\big(\mathrm{Obs}^{\mathrm{cl}}(Y), Q + \{S,-\}\big).$$

Cancellation requires the quartic trace identity

$$\mathrm{Tr}_{\mathfrak{g}}(X^4) = \lambda_{\mathfrak{g}}\big(\mathrm{Tr}_{\mathfrak{g}}(X^2)\big)^2.$$

The Deligne exceptional series appears because this identity is rigid there. On $K3 \times E$, integration over $K3$ contributes

$$\int_{K3} c_2(TK3) = 24,$$

so the anomaly coefficient is multiplied by the K3 Euler factor. The manuscript's Costello-comparison appendix and hCS chapters distinguish this genuine gauge-theoretic Green–Schwarz/axion cancellation from the algebraic shadow obstruction tower; the latter detects a residual quartic class but does not construct the cancellation. main

17. K3 protected Pfaffian: the missing operator

The scalar identity is not enough. The operator-level object must be a Dirac/Pfaffian element

$$\mathcal{D}_X \in \mathrm{End}_{\mathrm{ChirHoch}}\big(\mathcal{H}_X\big), \qquad X = K3 \times E,$$

whose protected Pfaffian is a section

$$\mathrm{Pf}^{\mathrm{prot}}(\mathcal{D}_X) \in H^0\big(\Lambda^{2,1}_{II}, \mathcal{L}_5 \otimes \nu_{\Delta_5}\big).$$

The desired identity is

$$\iota_{\mathrm{aut}}\left(\mathrm{Pf}^{\mathrm{prot}}(\mathcal{D}_X)\right) = \Delta_5.$$

The manuscript states this as one of the operator-level construction problems and identifies a resolution via BBDJS Pfaffian sections on the cosection-reduced d-critical heart with Kiem–Li cosection from the K3 holomorphic two-form, conditional on the named orientation and wall-crossing packages. main

The Hall side must produce an intertwiner

$$\mathfrak{I}_{\mathrm{Hall}} : \mathrm{CoHA}(K3) \longrightarrow \mathfrak{g}_{\Delta_5}$$

such that the square

$$\begin{tikzcd} \mathrm{ChirHoch}^\bullet(\mathrm{SpCh}\Phi^{FA}(K3)) \arrow[r, "\mathrm{Pf}^{\mathrm{prot}}"] \arrow[d, "\mathfrak{I}_{\mathrm{Hall}}"'] & H^0(\mathcal{L}_5) \arrow[d, "\mathrm{Borcherds}"] \\ \mathfrak{g}_{\Delta_5} \arrow[r, "\mathrm{den}"] & \Delta_5 \end{tikzcd}$$

commutes. Until this square is present, the Borcherds product is a scalar character, not the acting boundary object.

18. Chiral Springer and antipode from orientation reversal

The chiral Steinberg object should be

$$\mathfrak{St}_{\mathrm{ch}}(A) = L_A \times^h_{\mathcal{M}_{\mathrm{vac}}} L_A^\vee,$$

where L_A and $L_A^\vee$ are the Koszul-complementary Lagrangians in the shifted symplectic formal vacuum stack. The convolution product is

$$\mathcal{F} \star \mathcal{G} = p_{13,*}\big(p_{12}^*\mathcal{F} \otimes p_{23}^*\mathcal{G}\big)$$

on

$$L_A \times^h_{\mathcal{M}} L_A^\vee \times^h_{\mathcal{M}} L_A.$$

The antipode must be the orientation-reversal functor

$$\mathsf{S} : \mathfrak{St}_{\mathrm{ch}}(A) \to \mathfrak{St}_{\mathrm{ch}}(A), \qquad (x, y) \mapsto (y, x),$$

together with Verdier duality:

$$\mathsf{S}(\mathcal{F}) = \mathbb{D}_{\mathrm{Verdier}}\big(\iota^*\mathcal{F}\big).$$

The four antipode identities are

$$m(\mathsf{S} \otimes 1)\Delta = \eta\epsilon,$$

$$m(1 \otimes \mathsf{S})\Delta = \eta\epsilon,$$

$$\Delta\mathsf{S} = (\mathsf{S} \otimes \mathsf{S})\Delta^{op},$$

$$\epsilon\mathsf{S} = \epsilon.$$

The obstruction beyond class G is that orientation reversal must commute with the full higher MC tower:

$$\mathsf{S}(\Theta) = -\Theta + d_\Theta(\Xi)$$

for some homotopy Ξ. If the homotopy does not exist, one obtains an anti-equivalence of the associated graded, not an antipode of the full chiral Drinfeld double.

19. The concrete missing theorem statements

The manuscript should contain the following theorem fragments in this exact mathematical shape.

Boundary Kuranishi theorem.

$$F : E \to O, \quad F(p) = 0 \quad \Longrightarrow \quad K^{\mathrm{ex}}_{F,p} \simeq K_{\kappa_F},$$

with

$$\kappa_2 = F_2^C, \qquad \kappa_3 = F_3^C - \sum_{\text{cyc}} F_2^C(1, A^{-1}F_2^B).$$

Local bulk theorem.

$$HH^\bullet(K_{\kappa_F}) \simeq \mathcal{O}(d\,\mathrm{Crit}\langle\gamma, \kappa_F(u)\rangle).$$

Shifted RTT theorem.

$$Y_\mu^{RTT} = Y^{RTT}/I_\mu, \qquad \Delta I_\mu \subset I_\mu \otimes Y + Y \otimes I_\mu.$$

Rank-one test.

$$\mathrm{gr}\, Y_{\mu,A_1}^{RTT} \simeq \mathbb{C}[x,y,z]/(xy - z^m).$$

Baxter–Rees continuation.

$$R^{(\le w)}(u,\beta) = \sum_{d=0}^{D(w)} \beta^d R_d^{(\le w)}(u), \qquad d_{\Theta_0}\dot{\Theta} = 0.$$

Wild KZB composition.

$$\nabla \simeq d - d\Theta_\partial - \Lambda \frac{dq}{q}, \qquad S_{\ell_1,\ell_2} S_{\ell_2,\ell_3} S_{\ell_3,\ell_1} = 1.$$

Genus-two primitive.

$$P_2 = \frac{1}{12}\langle\sigma_2, \mathrm{ev}_{\Gamma_\flat}\rangle + \frac{1}{8}\langle\sigma_2, \mathrm{ev}_{\Gamma_\sharp}\rangle,$$

$$d_{\Theta^{(0)}} P_2 = D_1\Theta^{(1)} + \frac{1}{2}[\Theta^{(1)}, \Theta^{(1)}].$$

Anomaly-completed Steinberg.

$$B_\Theta = B\langle\eta\rangle, \qquad d\eta = \Theta, \qquad u = \eta^2,$$

$$\mathcal{G}_g[u^{-1}] \simeq \mathrm{Mat}_{2^g}(B_\Theta[u^{-1}]), \qquad \mathcal{G}_g/(u) \simeq (B_\Theta/(u)) \otimes \Lambda^{2g}.$$

Orbifold descent.

$$[\alpha_{\text{orb}}] \in H^3(BG; U(1)), \qquad [\alpha_{\text{orb}}] = 0 \iff \text{chain-level finite-orbifold descent}.$$

Protected K3 Pfaffian.

$$\iota_{\text{aut}}\left(\mathrm{Pf}^{\text{prot}}(\mathcal{D}_{K3\times E})\right) = \Delta_5.$$

Those are not embellishments. They are the missing local equations that make the maximalist monograph into mathematics rather than a sequence of identifications.

Sources

you started in the right direction, but you have left out IMMENSE detail. fill in all the gaps without repeating anything you stated earlier.

Thought for 10m 8s >

According to a document from 2026-06-05, the remaining missing body is concentrated in the chapters on spectral transport, celestial transfer, Yang–Mills boundaries, screening/mass-gap reduction, chiral CE factorisation, knot-theoretic bar integrals, non-principal W-algebras, and the six-dimensional hCS coefficient tower. main

Below is the next layer of actual mathematics, avoiding the earlier material.

1. Spectral discriminants must be modules, not numbers

The spectral discriminant

$$\Delta_A(x)$$

should never be used only as a scalar polynomial. The correct object is the cyclic transport module

$$M_A := M(p_A) = k[t]/(p_A(t)), \qquad p_A(t) = \Delta_A^{\mathrm{rev}}(t) = t^{d_A}\Delta_A(t^{-1}).$$

If

$$\Delta_A(x) = \prod_i (1-\lambda_i x)^{e_i},$$

then

$$p_A(t) = \prod_i (t-\lambda_i)^{e_i}.$$

Multiplication by t gives the transport operator

$$T_A : M_A \to M_A, \qquad [f(t)] \mapsto [tf(t)].$$

The point is that the branch divisor of Δ_A becomes a finite $k[t]$-module with subquotients. This is the algebraic object that remembers which spectral factors can be functorially transported through Drinfeld–Sokolov reduction, exact factorisation functors, embeddings, and quotient procedures. The manuscript explicitly says that the missing object is not Δ_A itself but the cyclic module cut out by the reversed polynomial, with divisors of Δ_A becoming actual subquotients. main

For a monic divisor $g \mid p$, define the divisor core

$$C_g(p) := \frac{p}{g} M(p) \subset M(p).$$

Then

$$C_g(p) \cong k[t]/(g).$$

For two divisors $g_1, g_2 \mid p$,

$$C_{g_1}(p) + C_{g_2}(p) = C_{\mathrm{lcm}(g_1,g_2)}(p),$$
$$C_{g_1}(p) \cap C_{g_2}(p) = C_{\gcd(g_1,g_2)}(p).$$

If $g_1 \mid g_2$, then

$$C_{g_1}(p) \subset C_{g_2}(p), \qquad C_{g_2}(p)/C_{g_1}(p) \cong k[t]/(g_2/g_1)$$

when g_1 and g_2/g_1 are coprime; otherwise the quotient carries a nontrivial extension class. That extension class is the Jordan-sector data.

The morphism spaces are controlled by gcd:

$$\mathrm{Hom}_{k[t]}\big(M(g_1), M(g_2)\big) \cong \ker\big(g_1(T) : M(g_2) \to M(g_2)\big),$$

and therefore

$$\dim_k \mathrm{Hom}_{k[t]}(M(g_1), M(g_2)) = \deg \gcd(g_1, g_2).$$

This is the exact algebra behind the phrase “maps are controlled by gcd.” It is also the missing replacement for any informal claim that a spectral factor gives a canonical projector.

There is a projector

$$e_g(t) \in k[t]/(p)$$

with

$$e_g^2 = e_g, \qquad \mathrm{im}\, e_g = C_g(p)$$

if and only if

$$\gcd(g, p/g) = 1.$$

On the repeated-root locus no such polynomial projector exists. The correct datum is the primary filtration

$$M\big(q^e\big) \supset qM\big(q^e\big) \supset q^2 M\big(q^e\big) \supset \cdots \supset q^{e-1} M\big(q^e\big) \supset 0,$$

with graded pieces

$$q^j M(q^e)/q^{j+1} M(q^e) \cong k[t]/(q).$$

This is where the monograph should place the Jordan geometry of degenerating spectral sheets. The manuscript explicitly states that projectors exist precisely on the coprime locus and that on repeated-root loci the invariant is a canonical subquotient plus an extension class. main

2. The Casimir recurrence module must be the reduced H^1

Let

$$Z_A = k[C_1, \dots, C_r, K]$$

be the commutative algebra generated by Casimirs and level. Let

$$u_A(n) := [H_n(\bar{B}(A))] \in K_0(Z_A\text{-mod}_{\text{fl}}) \otimes_{\mathbb{Z}} k.$$

The finite-character recurrence hypothesis is not decorative. It is the assertion that

$$p_A(\sigma)u_A = 0$$

for a minimal monic shift polynomial p_A, and that

$$p_A(t) = t^{d_A}\Delta_A(t^{-1}).$$

Then the intrinsic recurrence module is

$$R_A := k[\sigma] \cdot u_A \subset \left(K_0(Z_A\text{-mod}_{\text{fl}}) \otimes k\right)^{\mathbb{N}}.$$

The algebraic reduced first cohomology is

$$H^{1,\text{alg}}_{\text{red}}(A) := R_A^{\vee},$$

and the spectral transport endomorphism is

$$KS_A := \sigma^{\vee}.$$

The isomorphism that must be proved is

$$\eta_A : M(p_A) \ \ \tilde{} \ \ R_A, \qquad [t^j] \mapsto \sigma^j u_A.$$

This is a proof obligation, not notation: it is equivalent to showing that p_A is the minimal recurrence of the bar-cohomology class sequence. The manuscript states precisely this recurrence hypothesis and uses it to identify the recurrence module with the discriminant module. main

For $\mathfrak{sl}_2$, the scalar bar series in Riordan normalization satisfies

$$x(1+x)P_{\mathfrak{sl}_2}(x)^2 - (1+x)P_{\mathfrak{sl}_2}(x) + 1 = 0.$$

The branch divisor is

$$\Delta_{\mathfrak{sl}_2}(x) = (1+x)(1-3x),$$

hence

$$p_{\mathfrak{sl}_2}(t) = t^2 \Delta_{\mathfrak{sl}_2}(t^{-1}) = (t-3)(t+1),$$

and

$$M_{\mathfrak{sl}_2} = k[t]/\big((t-3)(t+1)\big).$$

The bar coefficients obey the polynomial recurrence

$$(n+1)a_n = (n-1)(2a_{n-1} + 3a_{n-2}),$$

whose nonconstant coefficient is the Orlik–Solomon/factorial collision contribution; it is not seen by invariant theory alone. The manuscript separates the Molien–Weyl/PBW constraint from the Arnold/Orlik–Solomon growth mode in exactly this calculation.

main

For a DS map

$$F : \mathfrak{g}_k \longrightarrow W_k(\mathfrak{g}, f),$$

the reduced spectral transport must factor through the common divisor

$$S_{\mathfrak{g},k}(x) = \gcd(\Delta_{\mathfrak{g}_k}(x), \Delta_{W_k(\mathfrak{g},f)}(x)),$$

with

$$s_{\mathfrak{g},k}(t) = S^{\mathrm{rev}}_{\mathfrak{g},k}(t).$$

The actual theorem should say:

$$F^{\vee}_{\mathrm{red}} : R_{\mathfrak{g}_k} \to R_{W_k(\mathfrak{g},f)}$$

factors uniquely as

$$R_{\mathfrak{g}_k} \twoheadrightarrow C_{s_{\mathfrak{g},k}} \longrightarrow R_{W_k(\mathfrak{g},f)}.$$

In the $\mathfrak{sl}_3/W_3$ denominator chart appearing in the manuscript,

$$S_{\mathfrak{g},k}(x) = 1 - 3x - x^2,$$

so

$$s_{\mathfrak{g},k}(t) = t^2 - 3t - 1,$$

and the missing defect factors are the two discarded linear factors

$$t-8, \qquad t-1.$$

This is the precise algebraic meaning of “DS preserves the common core but not the whole discriminant.” main

3. Relative Hochschild duality must be a Čech duality theorem

For an m-body boundary configuration, take a boundary simplex

$$\mathfrak{N} = \{M_S\}_{\varnothing \neq S \subset I}, \qquad |I| = m,$$

where M_S is the boundary category/algebra attached to the collision cluster S. Put

$$Z_S := HH^0(M_S), \qquad T_S := Z_S^\vee \otimes \omega_X.$$

The centre complex is

$$C_Z^p(\mathfrak{N}) = \bigoplus_{|S|=p+1} Z_S,$$

with differential

$$(d_Z z)_S = \sum_{i=0}^{p+1} (-1)^i \rho^*_{S\setminus\{s_i\},S}(z_{S\setminus\{s_i\}}).$$

The tangent complex is the reversed dual Čech complex

$$C_T^p(\mathfrak{N}) = \bigoplus_{|S|=m-p} T_S,$$

with differential

$$d_T = (-1)^{m-p} d_Z^\vee \otimes 1_{\omega_X}.$$

The missing theorem is the relative Hochschild duality

$$H^q(C_T^\bullet(\mathfrak{N})) \cong H^{m-1-q}(C_Z^\bullet(\mathfrak{N}))^\vee \otimes \omega_X.$$

Therefore the pure visible m-body coupling space is not an Ext group guessed from physics; it is

$$T_X^{1,[m]}(\mathfrak{N}) := H^0(C_T^\bullet(\mathfrak{N})) \cong H^{m-1}(C_Z^\bullet(\mathfrak{N}))^\vee \otimes \omega_X.$$

The manuscript gives this exact formula in the screened setting, where Z_S is first quotient by the screening ideal. main

The mixed Kodaira–Spencer class is the cocycle measuring the failure of the centre restrictions to be flat:

$$\theta_{S,T} \in HH^1(M_T, M_S), \qquad T \subset S,$$

and the higher-body obstruction is

$$\mathrm{ob}_{m+1}(\mathfrak{N}) = d_{\check{C}}\theta + \frac{1}{2}[\theta,\theta] \in H^2(C^\bullet_{\mathrm{relHoch}}(\mathfrak{N})).$$

For non-quadratic W-boundaries, the first nonzero term is not binary. The cubic and quartic pieces are

$$\theta_3 \in H^1(C^\bullet_T(\mathfrak{N}_3)), \qquad \theta_4 \in H^1(C^\bullet_T(\mathfrak{N}_4)),$$

and the first consistency condition is

$$d_{\check{C}}\theta_4 + \frac{1}{2}[\theta_3,\theta_3] = 0.$$

That is the higher-body replacement for the naive "bulk coupling" phrase.

4. Instanton completion is Novikov-completed Koszul duality

Let Γ be the effective instanton semigroup and let

$$\Lambda_{\geq 0} = \left\{ \sum_{\beta\in\Gamma} c_\beta Q^\beta : \#\{\beta : E(\beta) < R,\ c_\beta \neq 0\} < \infty \right\}.$$

An instanton-completed A_∞-chiral algebra has operations

$$m_k^\Lambda = \sum_{\beta\in\Gamma} Q^\beta m_{k,\beta},$$

with

$$E(\beta) \to +\infty$$

ensuring convergence. The completed bar differential is

$$D_A^\Lambda = \sum_{k,\beta} Q^\beta D_{m_{k,\beta}},$$

and the Maurer–Cartan equation is the energy-filtered hierarchy

$$\sum_{\substack{\beta_1+\beta_2=\beta \\ r+s+t=n}} \pm m_{r+1+t,\beta_1}(1^{\otimes r}, m_{s,\beta_2}, 1^{\otimes t}) = 0.$$

The Koszul dual is not

$$A^! \otimes \Lambda$$

unless finite energy propagation is proved. The correct definition is

$$A^!_\Lambda := \mathrm{Hom}^{\mathrm{cont}}_\Lambda\big(B_{\mathrm{ch}}(A)\hat{\otimes}\Lambda, \Lambda\big),$$

with continuous dual taken in the Q-adic topology.

The instanton correction to duality is the filtered map

$$\mathbb{D}_\Lambda B_{\mathrm{ch}}(A_\Lambda) \longrightarrow B_{\mathrm{ch}}(A^!_\Lambda)$$

and the obstruction at instanton class β is

$$o_\beta = \quad D_{m_{\bullet,\beta}} + \frac{1}{2} \sum_{\beta_1+\beta_2=\beta} [D_{m_{\bullet,\beta_1}}, D_{m_{\bullet,\beta_2}}] \quad \in H^2\,\mathrm{Coder}(B_{\mathrm{ch}}(A)).$$

This is the actual Novikov-completed Koszul-duality theorem that the instanton chapter should state.

5. Screenings must become a Hodge theorem and a gap inequality

A screening package is not a slogan about confinement. It is the following Hilbert complex.

Let H be a Hilbert completion of the state space, with vacuum projection P_0. Let

$$S_1, \ldots, S_r$$

be pairwise strongly commuting normal screening operators with common invariant core. Let

$$\epsilon_i, \iota_i : \Lambda^\bullet(\mathbb{C}^r) \to \Lambda^\bullet(\mathbb{C}^r)$$

satisfy

$$\epsilon_i \iota_j + \iota_j \epsilon_i = \delta_{ij}.$$

On

$$K_{\mathrm{scr}} := H \otimes \Lambda^\bullet(\mathbb{C}^r),$$

define

$$\partial_{\mathrm{scr}} = \sum_{i=1}^r \epsilon_i S_i, \qquad \partial^*_{\mathrm{scr}} = \sum_{i=1}^r \iota_i S_i^*.$$

Then the screening Laplacian is

$$L_{\mathrm{scr}} = \partial_{\mathrm{scr}}\partial^*_{\mathrm{scr}} + \partial^*_{\mathrm{scr}}\partial_{\mathrm{scr}} = \sum_{i=1}^r S_i^* S_i \otimes 1.$$

Consequently

$$\ker L_{\mathrm{scr}} = \bigcap_{i=1}^{r} \ker S_i \otimes \Lambda^\bullet(\mathbb{C}^r),$$

and degree zero harmonic states are exactly

$$H_{\mathrm{phys}} = \bigcap_i \ker S_i.$$

The manuscript states this screening Hodge theorem explicitly. main

The mass-gap transfer requires an actual inequality. The analytic hypothesis should be

$$\sum_i S_i^* S_i \geq \mu^2(1 - P_0)$$

as a quadratic-form inequality on the common core. Then

$$\mathrm{Spec}(L_{\mathrm{scr}}) \cap (0, \mu^2) = \varnothing,$$

and the physical screened cohomology has spectral gap at least μ.

The internal-screening domination criterion should be written:

$$\|P_{\mathrm{int}}\psi\|^2 \leq C \sum_i \|S_i \psi\|^2,$$

where P_{int} projects to the interacting non-vacuum sector. If this holds, the mass-gap problem reduces to proving the screening inequality for finitely many central screening operators.

The screened Čech duality is then obtained by quotienting every centre by the screening sequence:

$$Z_S^{\mathrm{vis}} = Z_S/(s_S), \qquad T_S^{\mathrm{vis}} = (Z_S^{\mathrm{vis}})^\vee \otimes \omega_X.$$

For a screened boundary simplex,

$$H^q(C^\bullet_{T,\mathrm{vis}}(\mathfrak{N})) \cong H^{m-1-q}(C^\bullet_{Z,\mathrm{vis}}(\mathfrak{N}))^\vee \otimes \omega_X.$$

That is the screened higher-body coupling theorem. main

6. Celestial transfer must be an obstruction tower, not a dictionary

Let C be the transferred boundary/celestial A_∞-algebra with a decreasing filtration

$$F^1 C \supset F^2 C \supset \cdots,$$

where F^r is the composite-degree filtration. The wedge quotient is

$$\mathrm{Wedge}(C) := F^1C/F^2C.$$

The transferred operations

$$\mu_n : C^{\otimes n} \to C$$

have associated graded components

$$\mathrm{gr}^r \mu_n : \mathrm{gr}^{i_1} C \otimes \cdots \otimes \mathrm{gr}^{i_n} C \to \mathrm{gr}^{i_1+\cdots+i_n+r} C.$$

The nonlinear obstruction classes are

$$\mathrm{Ob}_r(C) = [\mathrm{gr}^r \mu_{r+2}] \in H^1 \,\mathrm{Coder}\big(T^c(s^{-1}\,\mathrm{Wedge}(C))\big),$$

where the differential is

$$d_{\mathrm{wedge}} = [\mathrm{gr}^0 \mu_2, -].$$

The first nonlinearity theorem should say:

$$\mathrm{Ob}_1(C) = \big[\text{lowest nonlinear homogeneous part of the PVA } \lambda\text{-bracket}\big].$$

For W_3, this is the class of the composite field

$$\frac{16}{22+5c}\Lambda$$

inside $\{W_\lambda W\}$. The manuscript says the first celestial higher-spin correction for W_3 is controlled entirely by this composite term. main

The holomorphic BF ladder must be stated as a closed simplex formula. Let e_a denote a highest-weight generator of holomorphic degree a. The transferred n-ary product on the ladder has the form

$$\mu_n(e_{a_1},\ldots,e_{a_n}) = \left(\sum_{\text{comb } T} (-1)^{\epsilon(T)} I_T(a_1,\ldots,a_n)\right) e_{a_1+\cdots+a_n+n-2},$$

where

$$I_T(a_1,\ldots,a_n) = \int_{0<t_1<\cdots<t_{n-1}<1} \prod_{j=1}^{n-1} t_j^{A_j(T)}(1-t_j)^{B_j(T)}\,dt_j.$$

Equivalently,

$$I_T = \prod_{v\in V_{\mathrm{int}}(T)} \frac{\prod \Gamma(\alpha_{v,i}+1)}{\Gamma(\sum_i \alpha_{v,i}+|v|)}.$$

The Poisson BF odd ladder has slope two rather than slope one:

$$\deg w_1 = 2n-3$$

instead of

$$\deg w_1 = n - 2.$$

This extra degree is the cost of one holomorphic degree consumed by the Poisson bracket and one by the propagator; the manuscript explicitly marks this slope distinction. main

The Airy–Witt normal form should then appear only after the ladder is computed. The homogeneous first-order operators on $\mathbb{C}[t]$ are exactly

$$Q = a\, t^{d+1}\partial_t + b\, t^d,$$

and the odd highest-weight sector lands in the density-line representation of odd positive Witt modes. The manuscript proves this classification lemma and then identifies the transferred odd ladder with those first-order operators. main

For amplitudes, the missing theorem is not "bar complex resembles scattering." It is the identity

$$A_4^{\mathrm{bar}}(1^+, 2^-, 3^-, 4^+) = \int_{FM_4(\mathbb{C})} \omega_4^{(+,-,-,+)} \wedge \nu_4 = \frac{\langle 23\rangle^4}{\langle 12\rangle\langle 23\rangle\langle 34\rangle\langle 41\rangle}.$$

The weight form decomposes over the five planar binary trees:

$$\omega_4^{(+,-,-,+)} = \sum_{T\in PBT_4} \epsilon_T \prod_{e\in E(T)} d\log(z_{s(e)} - z_{t(e)}) \cdot K_T^{\bar\partial},$$

where $K_T^{\bar\partial}$ inserts the anti-holomorphic propagators for the negative-helicity states. The manuscript states the four-point MHV amplitude is reproduced by a bar-complex integral over $FM_4(\mathbb{C})$, with the degree-four weight form factorising over the five planar binary trees. main

7. The chiral Chevalley–Eilenberg algebra needs its generators and traces

For a smooth curve C, a finite-dimensional Lie algebra $\mathfrak{g}$, and a $\mathfrak{g}$-module M, the chiral CE complex is

$$C^\bullet_{\mathrm{ch}}(\mathfrak{g}, M)(C) = \Omega^{0,\bullet}(C) \otimes \mathrm{Sym}^\bullet(\mathfrak{g}^\vee[-1]) \otimes M.$$

Choose a basis T_a of $\mathfrak{g}$, structure constants

$$[T_b, T_c] = f^a{}_{bc} T_a,$$

and odd CE generators ξ^a. Then

$$d_{\mathrm{ch}} = \bar\partial_C + d_{\mathrm{CE}},$$

with

$$d_{\mathrm{CE}}\xi^a = -\frac{1}{2} f^a{}_{bc}\xi^b\xi^c,$$

and, for $m \in M$,

$$d_{\mathrm{CE}} m = \sum_a \xi^a \, T_a m.$$

This tridegree convention is essential:

$$\deg_{\mathrm{tot}} = \deg_{\bar\partial} + \deg_{\mathrm{CE}} + \deg_M .$$

The manuscript gives this definition and says the chiral CE algebra is only a local model for a Costello–Gwilliam observable factorisation algebra; the full cosheaf additionally requires observables on all opens, structure maps, and Weiss descent. main

On a formal disk with coordinate z, introduce modes

$$\xi^a_n := z^n \xi^a, \qquad m_n := z^n m.$$

Then

$$d_{\mathrm{CE}}\xi^a_n = -\frac{1}{2}\sum_{p+q=n} f^a{}_{bc}\xi^b_p\xi^c_q,$$

and

$$d_{\mathrm{CE}} m_n = \sum_{a,p+q=n} \xi^a_p \, T_a m_q.$$

The factorisation product is

$$(\alpha \otimes P(\xi) \otimes m)(z_1) \cdot (\beta \otimes Q(\xi) \otimes n)(z_2) = \alpha(z_1) \wedge \beta(z_2) \otimes P(\xi(z_1))Q(\xi(z_2)) \otimes (m \otimes n),$$

with singular corrections supplied by the chosen propagator.

The one-loop OPE correction must be written in motivic normalization. A typical coefficient has the form

$$\hbar \, \frac{\zeta(2)}{(2\pi i)^2} \, \frac{f^{ab}{}_d f^{dc}{}_e \, A^e(w)}{z-w}.$$

At genus one, the same coefficient is replaced by

$$\frac{\zeta(2)}{(2\pi i)^2}\Xi_2(\tau), \qquad \Xi_2(\tau) = \eta(\tau)^{-2}E_2(\tau),$$

transported by the Bergman propagator. The manuscript explicitly says every Feynman-traced OPE coefficient is tracked to a motivic representative $\zeta(k)/(2\pi i)^k$. main

For $\mathfrak{sl}_2$, the two-loop sunrise integrand should be printed:

$$I^{\mathrm{sun}}_{\mathfrak{sl}_2}(z,w) = \frac{1}{(2\pi i)^3} \oiiint \frac{\langle J^a(u)J^b(v)\rangle\langle J^c(u')J^d(w)\rangle\langle J^e(u)J^f(v)J^g(u')\rangle_{\mathrm{tree}}}{(u-v)^2(v-u')^2(u'-u)^2}\,du\,dv\,du'.$$

The coefficient is

$$\hbar^2 = -\frac{1}{8}$$

in the manuscript's $\mathfrak{sl}_2$ normalization. main

Wilson lines are factorisation modules:

$$W_V(\gamma) = P\exp\left(\int_\gamma A^a T_a^{(V)}\right),$$

and their OPE is

$$W_V(\gamma_1)W_W(\gamma_2) = R_{V,W}(z_{12})\,W_{V\otimes W}(\gamma_1\#\gamma_2).$$

The CE-to-factorisation theorem must identify

$$\int_{\mathbb{R}^2} \mathrm{Obs}^q_{\mathrm{hCS}} \simeq C^\bullet_{\mathrm{ch}}(\mathfrak{g},M)(C)$$

only after the Disk_3 action, Ayala–Francis–Tanaka Fubini pushforward, Weiss descent, and quantisation hypotheses are supplied. The manuscript names exactly these as the structural bridges. main

8. The six-dimensional hCS coefficient tower must be a graph complex

For six-dimensional holomorphic Chern–Simons on a Calabi–Yau threefold X, the field is

$$A \in \Omega^{0,1}(X, \mathrm{ad}\,P),$$

and in BV degree

$$A[1] \in \Omega^{0,\bullet}(X, \mathrm{ad}\,P)[1].$$

The action is

$$S_{\mathrm{hCS}}[A] = \int_X \Omega \wedge \left(\frac{1}{2}\langle A \wedge \bar\partial A\rangle + \frac{1}{6}\langle A \wedge [A \wedge A]\rangle\right).$$

The manuscript states this as the 6d hCS action and distinguishes it from ordinary three-dimensional Chern–Simons. main

The graph coefficient at loop order n should be

$$w_n = \sum_{\substack{\Gamma \text{ connected} \\ b_1(\Gamma)=n}} \frac{1}{|\operatorname{Aut}\Gamma|} \int_{\mathrm{Conf}_{V(\Gamma)}(X)} \mathrm{Tr}_{\mathfrak{g}} \quad \bigwedge_{e\in E(\Gamma)} P_e \prod_{v\in V(\Gamma)} I_v \quad .$$

The BV recursion is

$$d_{\mathrm{BV}} w_n + \frac{1}{2} \sum_{p+q=n} \{w_p, w_q\}_{\mathrm{BV}} = 0.$$

At one loop the anomaly is the quartic invariant

$$\mathcal{A}_1(X) = \int_X \Omega \wedge \mathrm{Tr}_{\mathrm{ad}}\big(A(\partial A)^3\big),$$

or invariantly the quartic Casimir obstruction

$$\mathrm{Tr}_{\mathfrak{g}}(X^4) - \lambda_{\mathfrak{g}}\big(\mathrm{Tr}_{\mathfrak{g}}(X^2)\big)^2.$$

The manuscript emphasizes that for complex dimension three the one-loop hCS anomaly is degree four and is not controlled by cubic-vanishing lists. main

At two loops the scalar channel is

$$w_2 \in \mathbb{Q} \cdot \zeta(3)$$

after the chosen motivic normalization.

At three loops the scalar channel decomposes as

$$w_3 = a\,\zeta(5) + b\,\zeta(3)^2.$$

At four loops the first genuinely non-Deligne graph-cohomology obstruction appears:

$$\xi_4 \in H^0(GC_2),$$

the “two-loop inside a two-loop” class. The manuscript says the scalar checks through loops $1, 2, 3$ are tested, while the all-loop extension is conjectural and at $n = 4$ the first non-obvious graph-cohomology obstruction appears. main

The all-loop target cannot be the bare weak Jacobi form. The proposed scalar comparison is

$$w_n^{\text{scalar}} \longleftrightarrow c_{1/\Delta_5}(q^n),$$

the q^n-Fourier–Jacobi coefficient of the Gritsenko–Nikulin Borcherds product. The chain-level comparison requires a finite Hall/BV source, an hCS-to-source chain map, root/parity data for $\mathfrak{g}_{\Delta_5}$, and maps to the Borcherds denominator lattice. The manuscript explicitly says the one-variable coefficients of $1/\Delta_5$ do not determine the chain-level BV coefficients. main

9. The Kontsevich integral must be a theorem of restricted operation spaces

For a Morse knot

$$K : S^1 \to \mathbb{C} \times \mathbb{R},$$

with height coordinate t, the Kontsevich integral is

$$Z(K) = \sum_{n \geq 0} \frac{1}{(2\pi i)^n} \int_{t_{\min} < t_1 < \cdots < t_n < t_{\max}} \sum_P \prod_{(i,j) \in P} \frac{d(z_i(t_i) - z_j(t_j))}{z_i(t_i) - z_j(t_j)} \cdot D_P.$$

The restriction of

$$FM_n(\mathbb{C}) \times \mathrm{Conf}_n(\mathbb{R})$$

to the graph of K is exactly the Kontsevich configuration space

$$\{t_1 < \cdots < t_n\} \times \{z_i = z_i(t_i)\}.$$

The bar weight form

$$\omega_n = \bigwedge_{(i,j)} d \log(z_i - z_j)$$

restricts to the Kontsevich integrand. This configuration-space identification is proved in the manuscript. main

The full equality

$$Z_{A_\infty}(K) = Z(K)$$

requires matching all signs, symmetry factors, associator normalisations, and framing corrections. The manuscript marks this full equality as conjectural beyond the verified abelian/genus-zero cases. main

The dictionary of relations must be explicit:

$$1T \quad \leftrightarrow \quad \mathfrak{g}\text{-invariance of } \Omega,$$

$$4T \quad \leftrightarrow \quad \text{two-chord Arnold cancellation},$$

$$IHX \quad \leftrightarrow \quad \omega_{12} \wedge \omega_{23} + \omega_{23} \wedge \omega_{31} + \omega_{31} \wedge \omega_{12} = 0,$$

$$STU \quad \leftrightarrow \quad \mathrm{Res}_{D_S}\, \omega_n = \omega_S \wedge \omega_{n/S}.$$

The manuscript says the dictionary accounts for all defining relations of the chord diagram algebra, with 1T, 4T, IHX, and framing independence accounted for. main

For non-formal targets such as Virasoro, the target cannot remain the chord-diagram algebra. The higher operations

$$m_k, \qquad k \geq 3,$$

produce higher-valent Jacobi diagrams. Thus the correct target is a completed graph algebra

$$\mathcal{A}_{\mathrm{Jac}}(A)$$

whose relations are generated not only by IHX but by the whole A_∞-Stokes hierarchy.

10. Non-principal W-algebras require orbit-sensitive bar data

The Bershadsky–Polyakov algebra is not just "another W_3." It is

$$W_3^{(2)} = H^0_{\mathrm{DS}}\big(V_k(\mathfrak{sl}_3), f_{\min}\big),$$

with minimal/subregular nilpotent input depending on convention. Its generator spectrum is

$$T \quad (h=2), \qquad J \quad (h=1), \qquad G^+ \quad (h=3/2), \qquad G^- \quad (h=3/2).$$

In the convention quoted in the manuscript,

$$c_{\mathrm{BP}}(k') = 2 - \frac{24(k'+1)^2}{k'+3}.$$

The same-family conductor identity is

$$c_{\mathrm{BP}}(k') + c_{\mathrm{BP}}(-k'-6) = 196.$$

The manuscript distinguishes this from the cross-family $N=2$ SCA/BP sum, which is not level-independent. main

The collision residues are

$$r_{TT}(z) = \frac{c/2}{z^3} + \frac{2T}{z},$$
$$r_{TG^\pm}(z) = \frac{(3/2)G^\pm}{z},$$

and

$$r_{G^+G^-}(z) = \frac{\alpha}{z^2} + \frac{\beta T + \gamma :G^+G^-:}{z}.$$

The G^+G^--channel has an even collision pole z^{-2}, because the original OPE has a triple pole and the bar logarithmic kernel absorbs one pole. The manuscript explicitly identifies this as the parity signature of half-integer generators. main

The shadow structure is orbit-sensitive:

$$\text{overall class}(W_3^{(2)}) = M$$

because the Virasoro subsector has an infinite tower, but the $G^{\pm}$-sector has finite depth:

$$m_3^{G^{\pm}} \neq 0, \qquad m_4^{G^{\pm}} = 0, \qquad m_k^{G^{\pm}} = 0 \quad (k \geq 4),$$

by residual $\mathfrak{sl}_2$-Jacobi cancellation. The manuscript states exactly that the $G^{\pm}$-sector behaves like class C inside an overall class M algebra. main

The linear coproduct is primitive on free generators but not on composites:

$$\Delta_z(G^{\pm}) = G^{\pm} \otimes 1 + 1 \otimes G^{\pm},$$

$$\Delta_z(J) = J \otimes 1 + 1 \otimes J,$$

but for the Sugawara stress tensor,

$$\Delta_z(T) = T \otimes 1 + 1 \otimes T + \Omega_{JJ}\, J \otimes J + \cdots .$$

The manuscript resolves this point: generator-primitivity survives, but composite-primitivity can fail because surviving weight-one currents contribute split-Casimir cross terms. main

For the $N = 2$ superconformal algebra, the OPEs must be printed, because they are the first mixed-parity bar test:

$$J(z)J(w) \sim \frac{c/3}{(z-w)^2},$$

$$J(z)G^{\pm}(w) \sim \pm\frac{G^{\pm}(w)}{z-w},$$

$$G^{+}(z)G^{-}(w) \sim \frac{c/3}{(z-w)^3} + \frac{J(w)}{(z-w)^2} + \frac{T(w) + \frac{1}{2}\partial J(w)}{z-w}.$$

The desuspended generators $s^{-1}G^{\pm}$ are parity-shifted, so the bar sign in

$$d_B[s^{-1}G^{+}|s^{-1}G^{-}]$$

is not the bosonic sign. This is where Appendix A's mixed-parity brace signs must be instantiated in the main text. The manuscript says this is the first concrete example where odd desuspensions affect the bar differential. main

Spectral flow should act on modes by

$$J_n \mapsto J_n + \frac{c}{3}\eta\, \delta_{n,0},$$

$$L_n \mapsto L_n + \eta J_n + \frac{c}{6}\eta^2 \delta_{n,0},$$

$$G_r^\pm \mapsto G_{r\pm\eta}^\pm.$$

The Koszul dual must intertwine this action. Thus the line-category theorem must include

$$\mathsf{SF}_\eta(A^!_{\mathrm{line}}) \simeq (\mathsf{SF}_\eta A)^!_{\mathrm{line}},$$

not merely a statement that spectral flow exists.

For $\mathfrak{sl}_4$, the three intermediate nilpotent orbits have different ordered-bar spectra:

$$\begin{aligned} (3,1): &\quad \{1,2,2,2,3\}, \\ (2,2): &\quad \{1,1,1,2,2,2,2\}, \\ (2,1,1): &\quad \{1,1,1,1,3/2,3/2,3/2,3/2,2\}. \end{aligned}$$

The manuscript records these as the first place where the nilpotent poset produces distinct ordered-bar complexes beyond the affine/principal lineage. main

11. Supersymmetric indices should be annular bar traces

The annular bar complex for an $N=2$ boundary algebra must carry three commuting operators:

$$L_0, \qquad J_0, \qquad (-1)^F.$$

The elliptic genus is the supertrace

$$\mathrm{Ell}_A(\tau,z) = \mathrm{Str}_{H^\bullet(B^{\mathrm{ann}}(A))}\left((-1)^F y^{J_0} q^{L_0-c/24}\right), \qquad q=e^{2\pi i\tau},\ y=e^{2\pi i z}.$$

The annular differential must commute with the index operator:

$$[d_{\mathrm{ann}}, L_0]=0, \qquad [d_{\mathrm{ann}}, J_0]=0, \qquad [d_{\mathrm{ann}}, (-1)^F]=0.$$

The weak Jacobi property should be derived from two annular facts:

$$\mathrm{Ell}_A(\tau, z+\lambda\tau+\mu) = e^{-2\pi i m(\lambda^2\tau+2\lambda z)}\,\mathrm{Ell}_A(\tau,z),$$

and

$$\mathrm{Ell}_A\left(\frac{a\tau+b}{c\tau+d}, \frac{z}{c\tau+d}\right) = (c\tau+d)^w e^{2\pi i m\frac{cz^2}{c\tau+d}}\,\mathrm{Ell}_A(\tau,z).$$

The index m is fixed by the J_0J_0-annular curvature:

$$m = \frac{1}{2}\operatorname{Res}_{z=w} J(z)J(w).$$

The manuscript has an annular-bar supersymmetric-index section for $N=2$ elliptic genus, weak Jacobi form property, $N=4$ index, and BPS spectrum from bar cohomology. main

For small $N = 4$, with $SU(2)_R$ currents J^a at level k,

$$c = 6k,$$

and the refined index must be

$$\mathcal{I}_{N=4}(\tau, z, u) = \mathrm{Str}\left((-1)^F q^{L_0 - c/24} y^{J_0^3} u^{K_0}\right),$$

where K_0 is the outer $SU(2)$ Cartan. The $SU(2)_R$ current algebra forces extra cancellations in the shadow tower; the manuscript explicitly says the small $N = 4$ algebra partially tames higher-degree operations through R-symmetry constraints. main

12. Chiral Springer resolution must resolve the singular Steinberg object

The singular chiral Steinberg object should be

$$\mathfrak{St}_{\mathrm{ch}}(A) = L_A \times^h_{\mathcal{M}_A} L_A^\vee.$$

Its singular locus is controlled by the failure of the two Lagrangians to meet cleanly:

$$\mathrm{Sing}\, \mathfrak{St}_{\mathrm{ch}}(A) = \mathrm{Spec}\, \mathrm{Tor}^{>0}_{\mathcal{O}_{\mathcal{M}_A}}(\mathcal{O}_{L_A}, \mathcal{O}_{L_A^\vee}).$$

For type A_1, the local model should be

$$xy = z^2$$

or its q-deformed/Kleinian variant, with resolution

$$\mathfrak{St}_{\mathrm{ch}} = \mathrm{Bl}_{(x,z)}\, \mathrm{Spec}\, k[x, y, z]/(xy - z^2).$$

For type A_n, the iterated resolution is indexed by collision strata:

$$\mathfrak{St}^{A_n}_{\mathrm{ch}} = \mathrm{Bl}_{D_{[1,n]}} \cdots \mathrm{Bl}_{D_{[i,j]}} \cdots \mathfrak{St}^{A_n}_{\mathrm{ch}},$$

where

$$D_{[i,j]}$$

is the diagonal collision stratum corresponding to the root interval

$$\alpha_i + \cdots + \alpha_j.$$

The convolution product is

$$\mathcal{F} \star \mathcal{G} = p_{13,*}\big(p_{12}^* \mathcal{F} \otimes p_{23}^* \mathcal{G}\big),$$

on

$$\mathfrak{St}_{\mathrm{ch}} \times_{L_A} \mathfrak{St}_{\mathrm{ch}}.$$

The resulting convolution algebra should match the Yangian/shifted Yangian after applying equivariant K-theory or Borel–Moore homology:

$$K^{G\times\mathbb{C}^\times}(\mathfrak{St}_{\mathrm{ch}}) \simeq Y_\hbar(\mathfrak{g})$$

on the affine lineage, with the shifted version obtained by imposing boundary/defect coweight conditions. The manuscript's frontier explicitly identifies chiral Springer resolution, type A_1, type A iterated blowups, and the convolution algebra/Yangian comparison as a remaining endpoint. main

13. The Yang–Mills boundary lane needs central formality

Let the twisted Yang–Mills boundary algebra be

$$A_{\mathrm{YM}} = C^\bullet_{\mathrm{ch}}(\mathfrak{g}, \mathcal{O}_{\mathrm{matter}})$$

with differential

$$d_{\mathrm{YM}} = \bar\partial + d_{\mathrm{CE}} + \iota_\mu,$$

where

$$\mu : \mathcal{O}_{\mathrm{matter}} \to \mathfrak{g}^\vee$$

is the moment map. The mixed coupling is the Hochschild class

$$\Theta_{\mathrm{mix}} \in HH^2(A_{\mathrm{YM}}, A_{\mathrm{YM}})$$

represented locally by

$$\Theta_{\mathrm{mix}} = \sum_a J^a_{\mathrm{gauge}} \otimes \mu_a^{\mathrm{matter}}.$$

Central formality is the statement that the map

$$Z(A_{\mathrm{gauge}})\otimes Z(A_{\mathrm{matter}}) \to HH^\bullet(A_{\mathrm{YM}})$$

is a quasi-isomorphism after restricting to the centre-generated subcomplex. Equivalently, the cone

$$\mathrm{Cone}\left(Z(A_{\mathrm{gauge}})\otimes Z(A_{\mathrm{matter}}) \to HH^\bullet(A_{\mathrm{YM}})\right)$$

has no degree 0 or 1 cohomology on the mixed-coupling support.

The deformation equation is

$$d\Theta_{\mathrm{mix}} + \frac{1}{2}[\Theta_{\mathrm{mix}}, \Theta_{\mathrm{mix}}] = 0.$$

The first obstruction is

$$[\Theta_{\mathrm{mix}}, \Theta_{\mathrm{mix}}] = \sum_{a,b}[J^a, J^b] \otimes \mu_a\mu_b + \sum_{a,b} J^a J^b \otimes \{\mu_a, \mu_b\}.$$

Gauge invariance cancels this exactly when

$$\{\mu_a, \mu_b\} = f_{ab}{}^c\mu_c.$$

Thus the Yang–Mills boundary theorem is a moment-map Jacobi theorem at Hochschild degree two, not a vague gauge-theory analogy. The manuscript's Yang–Mills chapter is organized around the twisted YM boundary package, central formality, and mixed-coupling theorems. main

14. The missing one-line replacements

Replace

$$\text{“the discriminant controls transport”}$$

by

$$H^{1,\mathrm{alg}}_{\mathrm{red}}(A) = \left(k[t]/(\Delta_A^{\mathrm{rev}})\right)^\vee, \qquad KS_A = t^\vee.$$

Replace

$$\text{“DS preserves spectral data”}$$

by

$$R_{\mathfrak{g}_k} \twoheadrightarrow C_{\gcd(\Delta_{\mathfrak{g}}, \Delta_W)^{\mathrm{rev}}} \hookrightarrow R_{W_k(\mathfrak{g},f)}.$$

Replace

$$\text{“higher-body coupling”}$$

by

$$T_X^{1,[m]}(\mathfrak{N}) = H^0(C_T^\bullet(\mathfrak{N})) \cong H^{m-1}(C_Z^\bullet(\mathfrak{N}))^\vee \otimes \omega_X.$$

Replace

$$\text{“screening gives confinement”}$$

by

$$L_{\mathrm{scr}} = \sum_i S_i^* S_i, \qquad \sum_i S_i^* S_i \geq \mu^2(1 - P_0).$$

Replace

$$\text{“celestial correction”}$$

by

$$\mathrm{Ob}_r = [\mathrm{gr}^r\,\mu_{r+2}] \in H^1\,\mathrm{Coder}\big(T^c(s^{-1}\,\mathrm{Wedge}(C))\big).$$

Replace

$$\text{“hCS coefficients match Borcherds”}$$

by

$$d_{\mathrm{BV}} w_n + \frac{1}{2}\sum_{p+q=n}\{w_p, w_q\}_{\mathrm{BV}} = 0, \qquad w_n \in \mathbb{Q}\langle \mathrm{MZV}_{\le 2n+1}\rangle,$$

plus the finite Hall–Borcherds comparison map.

Replace

$$\text{“bar complex gives knot invariants”}$$

by

$$FM_n(\mathbb{C}) \times \mathrm{Conf}_n(\mathbb{R})|_K = \{t_1 < \cdots < t_n,\ z_i = z_i(t_i)\},$$

$$\bigwedge d\log(z_i - z_j)\ \ _K = \text{Kontsevich integrand.}$$

Replace

$$\text{“BP is class M”}$$

by the sector statement

$$\begin{aligned} r_{TT}(z) &= \frac{c/2}{z^3} + \frac{2T}{z}, \\ r_{TG^\pm}(z) &= \frac{3G^\pm}{2z}, \\ r_{G^+G^-}(z) &= \frac{\alpha}{z^2} + \frac{\beta T + \gamma :G^+G^-:}{z}, \end{aligned}$$

with

$$m_3^{G^\pm} \neq 0, \qquad m_{\ge 4}^{G^\pm} = 0, \qquad m_k^{TT} \neq 0 \text{ infinitely often.}$$

Replace

$$\text{“supersymmetric index”}$$

by

$$\mathrm{Str}_{H^\bullet(B^{\mathrm{ann}}(A))}(-1)^F y^{J_0} q^{L_0 - c/24}.$$

Replace

$$\text{“chiral Springer gives Yangian”}$$

by

$$K^{G\times\mathbb{C}^\times}\left(\mathrm{Bl}_{D_{\alpha_1+\cdots+\alpha_n}}\cdots\mathrm{Bl}_{D_{\alpha_i+\cdots+\alpha_j}}\mathfrak{St}_{\mathrm{ch}}\right)\simeq Y_\hbar(\mathfrak{g}).$$

That is the next missing stratum: divisor-core spectral transport, Čech relative Hochschild duality, Novikov/screening analytic control, celestial obstruction classes, chiral CE Feynman traces, 6d hCS graph coefficients, Kontsevich restriction, and non-principal W-sector arithmetic.